SANOS-Evolve: A Discrete Stochastic-Local-Volatility Model for European-Option Smile Dynamics

From arbitrage-free marginals to the skew-stickiness ratio, with SPX as the worked example

Research Note

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Abstract

An arbitrage-free set of option marginals fixes an underlying's local-volatility content, but not how its smile moves when the spot moves. The models that get the movement right are either parametric in the smile, and so cannot reprice it exactly (stochastic volatility), or exact but expensive to calibrate (stochastic-local volatility).

We introduce SANOS-Evolve, a discrete stochastic-local-volatility model that keeps these two problems separate. A static layer fixes strictly arbitrage-free marginals with the SANOS linear program. A dynamic layer adds a fitted martingale transition kernel between consecutive marginals, carrying a latent volatility regime with two mean-reversion timescales. The marginals stay fixed while the dynamics are calibrated, and every admissible kernel is free of static arbitrage by construction (existence by convex order, Strassen).

Because every object is a Gaussian mixture in log-moneyness, prices, Greeks, and the variance-index smile are closed-form finite sums of Black–Scholes terms. The kernel is translation-invariant in log-spot, which has two consequences. First, the marginals are matched by a deterministic leverage overlay—a discrete Gyöngy identity—rather than a fitted projection. Second, the skew-stickiness ratio (SSR), the standard measure of how the smile rolls with spot, is an exact closed-form realised covariance, and the two timescales fit its term structure. The kernel's vol-of-vol is identified from a forward-variance readout (VIX for SPX; the option strip's own curvature otherwise), not by jointly calibrating a volatility-index market, so the construction stays portable across asset classes.

From one calibrated object a desk obtains the vanilla and forward-volatility surface, exotics, de-evented smiles, and SSR-consistent hedge ratios. SPX is the worked example; in a historical replay the SSR-consistent delta cuts realised hedging variance 27–47% below the Black delta.

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1 Introduction

An equity-index volatility surface poses two distinct problems. The *static* problem is to fit, at each listed maturity, a risk-neutral marginal that reprices the quoted vanillas within bid–ask and is free of butterfly and calendar arbitrage. The *dynamic* problem is to say how that surface moves when the spot moves. This smile dynamics governs forward-starting products, path-dependent exotics, and hedge ratios, and solving the static problem does not solve it.

Statics: marginals are local volatility. By Dupire's formula [2], a complete set of arbitrage-free marginals across maturities *is* the local-volatility content of the surface. The local variance $\sigma^2_{\mathrm{LV}}(K,T) = \partial_T C/(\frac{1}{2}K^2\partial_{KK}C)$ depends only on the marginals, and SANOS [1] solves this static problem: it writes call prices as weighted sums of Black–Scholes basis functions and fits the weights by a linear program (LP) with hard bid–ask and calendar constraints. The output is a smooth, strictly arbitrage-free marginal at each maturity, and with it a discrete Markov martingale whose canonical transition operator is of the rigid local-volatility type.

Dynamics: three families, and a gap. The local-volatility model implied by those marginals [4, 5], however, has the *wrong* dynamics. Its forward smiles flatten too quickly with maturity, and its skew-stickiness ratio—the SSR, the normalised co-movement of at-the-money volatility with spot (Section 6)—is a rigid consequence of the static skew rather than a free target. Stationary local volatility gives the linear limit $\mathrm{SSR} \to 2$; calibrated to the empirical $\tau^{-1/2}$ skew it is pinned near the Doeff–Kamal value $\mathrm{SSR} \approx \frac{3}{2}$ [32]. The empirical SPX term structure instead runs 1.9, 1.45, 1.0 from one week to two years [22, 23].

Stochastic volatility puts the randomness in the volatility itself and produces genuine smile dynamics, but it pays twice. Its parametric marginals cannot reprice the vanilla smile exactly—Heston [9] and SABR [10] miss the short-dated SPX skew—and its dynamics are also rigid: calibrated to the same smile, the whole family (Heston, two-factor and rough Bergomi [23]) clusters on one SSR term structure, and that clustering deviates from the empirical one [27, 25].

Stochastic-local volatility (SLV) restores the exact smile by multiplying a stochastic backbone by a local-volatility leverage, calibrated through Gyöngy's theorem [6, 7, 8]. The cost is that the leverage is a fixed point, solved by particle or PDE methods [15, 14]—more recently by optimal transport [16] or neural SDEs [17]—so there is no closed-form transition kernel and the Greeks carry Monte-Carlo noise. And because the leverage is spent matching the smile, the SSR is left to the backbone and inherits its rigidity (Section 9 makes the comparison precise).

So the field offers three incomplete options: exact but static (local volatility), dynamic but parametric (stochastic volatility), and exact and dynamic but intractable (SLV). Missing is a kernel that is at once tractable (closed-form), data-driven (fits any arbitrage-free smile exactly), and dynamically controllable (the SSR a free target). SANOS-Evolve is that kernel.

SANOS-Evolve: marginals plus a fitted kernel. SANOS supplies arbitrage-free marginals $\{\mu_j\}$ but pairs them with a rigid canonical transition operator, and its authors leave the controllable discrete-time dynamics as an open direction [1]. SANOS-Evolve keeps the marginals and replaces that operator with a fitted martingale transition kernel $\mathcal{K}_j$ between consecutive marginals, carrying a latent volatility regime with two mean-reversion timescales:

$$\underbrace{\{\mu_j\}}_{\substack{\text{SANOS marginals} \\ \text{(data-driven statics)}}} \quad + \quad \underbrace{\{\mathcal{K}_j\}}_{\substack{\text{martingale kernel} \\ \text{(controllable dynamics)}}} \quad = \underbrace{\text{closed-form discrete SLV}}_{\text{for any European option.}}$$

We fix the marginals from quotes and fit the kernel between them, rather than postulate a process and back out a fit. The SANOS no-arbitrage structure is therefore preserved exactly while the dynamics are calibrated. Two properties make this work.

The kernel is expressive enough. A Gaussian mixture placed in the *kernel*—not, as in Brigo–Mercurio [13], in the marginal of a shared-forward diffusion—is dense in Wasserstein-1 on each fibre among the martingale kernels consistent with the marginals (Appendix C). It can carry any admissible dynamics, so the marginal match is never the binding constraint. We nonetheless keep a parsimonious two-timescale form, which pins the dynamics to interpretable regimes rather than leaving them underdetermined.

We know what the kernel must fit. The smile's motion, as distinct from its shape, decomposes into a small basis, ordered by the order each term takes in the vol-of-vol expansion [23]: a *leverage* (spot–vol coupling, first order), a *vol-of-vol* (second order), and a *vol-of-vol skew* (third order). Each is carried across maturity by the volatility process's timescales, and at the short end by its roughness. The SSR is the leading coordinate—the leverage normalised by the static skew, $\mathrm{SSR}_T = -\beta_T\sqrt{T}/\mathcal{S}_T$ with $\beta_T = \mathrm{d}\widehat{\sigma}_{\mathrm{ATM}}/\,\mathrm{d}\log S$—so fitting it decouples the dynamic response from the static shape. SANOS-Evolve calibrates the first- and second-order dynamics (the SSR, a VIX-informed vol-of-vol readout, and the two timescales) separately from the statics; higher-order terms are named extensions (Section 6). The variance index enters as information rather than a target: a soft penalty that *identifies* the kernel's vol-of-vol. This is what separates the construction from joint SPX/VIX calibration: dispersion-constrained bridges [30] and the perturbed-OT coupling [31] both fit the two smiles as hard targets. A hard VIX target presumes a volatility-index *options* market deep enough to calibrate across strikes—which among European underlyings effectively means SPX alone; VSTOXX and VXN options exist but trade an order of magnitude thinner, and no single name, FX pair, or crypto has a VIX-comparable one. A forward-variance *readout*, by contrast, needs only the index level or futures where they exist, or the option strip's own curvature otherwise, so it identifies the vol-of-vol for far more underlyings (Table 2). Reading it is therefore the general construction, of which the SPX+VIX case is the sharpest instance—the price is honest, more portable but less tightly identified precisely where a volatility index does exist (Section 6). The basis, with the kernel knobs (Section 4) that carry each characteristic, is:

characteristic	order	smile signature	in the kernel
leverage (the SSR)	1st	skew + its stickiness	$\lambda_{\mathrm{skew}}, \lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$
vol-of-vol	2nd	convexity, forward-smile spread, VIX level	$\nu_{\mathsf{F}}, \nu_{\mathsf{S}}, \nu_{\mathsf{L}}$
vol-of-vol skew	3rd	VIX-smile slope, deep-wing asymmetry	— (extension)
timescales	—	maturity-decay of the above	$\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$
roughness H	—	short-T blow-up of skew/SSR	$H{=}\frac{1}{2}$ fixed (extension)

Figure 1 places SANOS-Evolve among transition-kernel families by three properties: the explicitness of the kernel (horizontal), the dynamic freedom (vertical), and the exactness of the marginal fit (fill).

Contributions.

1. **A discrete SLV that decouples statics from dynamics (Sections 2–3).** A convex LP fixes strictly arbitrage-free marginals; a separately fitted martingale kernel carries the dynamics, and the marginals stay invariant when the dynamic targets are re-tuned. Any admissible chain is free of static arbitrage, with existence guaranteed by convex order (Strassen [20]). There is no leverage fixed point and no existence cliff, unlike classical SLV.

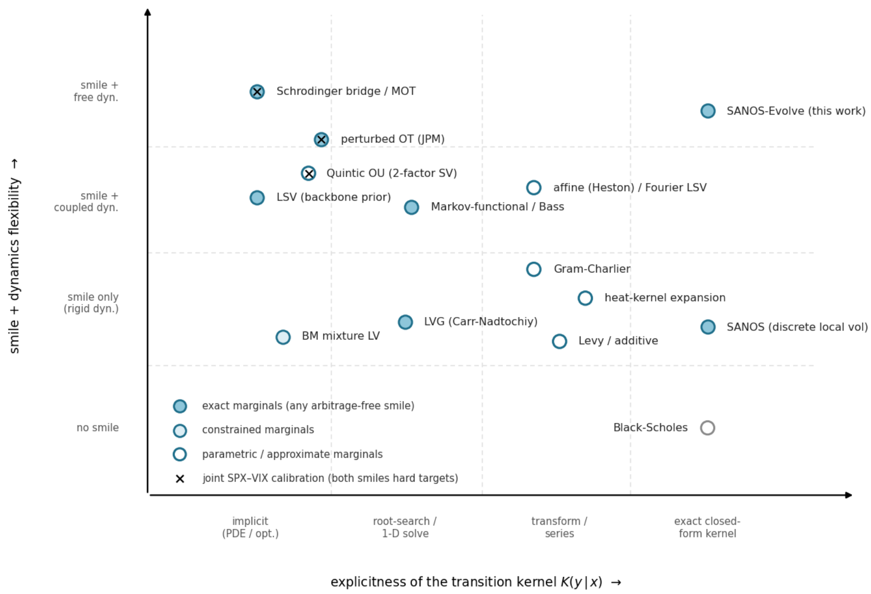

Figure 1: **The transition-kernel landscape.** Explicitness of the one-step kernel $K(y \mid x)$ (horizontal), smile-and-dynamics freedom (vertical), exactness of the marginal fit (fill); tiers and legend on the plot (Black–Scholes: no smile, fit n/a, grey). In the rightmost column, SANOS-Evolve replaces SANOS's rigid canonical coupling with a free-dynamics Gaussian-mixture kernel (Appendix C). A black cross marks joint SPX–VIX calibration (both smiles as hard targets)—the programme the VIX-as-readout lane of Section 1 stays out of. *Abbreviations:* MOT/OT, (martingale) optimal transport; LSV, stochastic-local volatility; LVG, local variance gamma; BM, Brigo–Mercurio; LV, local volatility; SV, stochastic volatility; OU, Ornstein–Uhlenbeck.

2. **The SSR as an exact closed-form realised covariance, free of the smile (Sections 4, 6).** A translation-invariant two-timescale kernel places the spot–vol leverage in the return innovation around fast and slow regimes. Translation invariance then makes the SSR an exact realised covariance of return and regime—not a spot-derivative or a soft penalty—and the two timescales give its term structure, fit separately from the smile. This is the freedom the calibrated stochastic-volatility family lacks.

3. **Closed-form throughout, hence deterministic (Sections 5, 6).** Prices, Greeks, the variance-index smile, and the SSR and forward-variance readouts are all finite Black–Scholes mixtures. Hedge ratios are therefore reproducible and noise-free, where particle SLV carries Monte-Carlo noise and PDE SLV carries grid error. The vol-of-vol is identified from a variance-index (VIX) soft target rather than by jointly calibrating a volatility-index market.

4. **Scheduled-event de-eventing of the spot–vol dynamics (Section 7).** A clean-flank bridge splits a scheduled event into two closed-form pieces: a marginal jump density, and—the part the literature leaves open—an event-conditional SSR that records whether volatility jumps *with* the spot at the release.

Related work. Figure 1 places SANOS-Evolve among transition-kernel families. The closest dynamic relative is regime switching: Jourdain and Zhou [18] prove existence of a calibrated regime-switching model, a special case of local-stochastic volatility in which a finite-state process, not the spot, carries the randomness. Our latent regime is finite-state in the same spirit, but two-timescale and calibrated to the SSR term structure. The fast/slow structure itself is standard multi-factor stochastic volatility in the Bergomi lineage [22]. The Quintic OU model [28]—the same two-OU state under a polynomial volatility map—likewise steers the SSR separately: even its joint SPX/VIX calibration needs an SSR band penalty, independent evidence that the smiles do not pin the SSR. Its polynomial map buys one closed-form readout, the VIX functional; we keep the exponential map and discretise the *state* (Section 4), which buys the whole kernel. That forward implied volatility needs a transition density, not merely marginals, is the lesson of [19].

We build on martingale optimal transport [20, 21]: we adopt its admissible-set constraint (with Strassen existence) but replace its objective, fitting an economically structured kernel to dynamic targets rather than extremising an exotic's price. A neural mixture-density surrogate [29] reuses the same Gaussian-mixture and CDF-loss machinery for the opposite, simulation-surrogate objective. Scheduled-event stripping, finally, is established at the marginal level—a discrete event-variance step [34, 35], or a clean-surface-plus-jump decomposition from non-event-spanning expiries [37, 36]—but none of it carries the event's spot–vol dynamics. That gap is what our kernel de-eventing (Section 7) fills, an open extension that a full-surface ultra-short model flags without building [38].

Scope. SANOS-Evolve applies to any European-option underlying. The construction needs only European exercise (so the terminal marginal is read from prices), a liquid surface, and a forward-variance readout for the vol-of-vol. We take SPX as the worked instance, where a liquid VIX gives the sharpest readout; equity indices follow directly, and FX, crypto, and European single names reuse the same machinery with their own readout and SSR (Section 6). Throughout, we use a readout to *identify* vol-of-vol, not to jointly calibrate a volatility-index options market—the route of the perturbed-OT desk method [31]. Generality here is structural, a property of the construction rather than an empirical claim across asset classes: the study is SPX. The daily framework is a methods contribution with ground-truth validation and a real-data calibration (Section 8); a brief intraday/0DTE outlook closes the discussion (Section 9).

2 The static layer: SANOS marginals as local volatility

Fix a pricing date and maturities $T_0 < T_1 < \cdots < T_M$ with forwards F_j and discount factors D_j. Work in log-moneyness $z = \log(S/F)$. SANOS fits, at each T_j, an arbitrage-free density as a Gaussian mixture in z,

$$\rho_{S,T_j}(z) = \sum_i q_j^i\,\mathcal{N}\big(z;\mu_j^i,(\sigma_j^i)^2\big), \qquad q_j^i \geq 0,\ \sum_i q_j^i = 1, \tag{1}$$

by a linear program over the weights q_j subject to hard bid–ask boxes, forward (martingale) normalisation, and calendar constraints [1]. We denote the resulting marginal law $\mu_j = \mathrm{Law}(S_{T_j})$. By Breeden–Litzenberger [3], (1) is butterfly-arbitrage-free iff $\rho_{S,T_j} \geq 0$, which holds by construction, and the family $\{\mu_j\}$ is calendar-arbitrage-free iff it is increasing in convex order, which the SANOS LP enforces. By Dupire [2], $\{\mu_j\}$ *is* the local-volatility content of the surface.

Two structural facts make the whole construction closed-form. First, in log-coordinates every object we use is a Gaussian mixture. Second, Black–Scholes is not a modelling assumption but a computational identity: the call price under a single Gaussian log-return is the moment-generating function of a truncated Gaussian,

$$\mathbb{E}\big[(S-K)^+\big] = F\,\Phi(d_1) - K\,\Phi(d_2), \qquad d_{1,2} = \frac{\log(F/K) \pm \frac{1}{2}\tau^2}{\tau}, \tag{2}$$

with $\tau = \sigma\sqrt{T_j}$ the total volatility. The price of any claim is then a *pairing* of the marginal with its payoff; for the mixture (1) the call is a finite weighted sum of Black–Scholes evaluations,

$$C_j(K) = \langle \mu_j, (\cdot - K)^+ \rangle = \sum_i q_j^i \ \mathrm{BS}\big(F_j^i, K, \sigma_j^i\big), \qquad F_j^i = F_j\, e^{\mu_j^i + \frac{1}{2}(\sigma_j^i)^2}, \tag{3}$$

with no quadrature, no PDE, and no Monte Carlo, and likewise for digitals and any payoff with a closed-form Gaussian expectation. Because call payoffs separate measures (Breeden–Litzenberger), this pairing is a faithful coordinate for μ_j: the mixture density makes the marginal matchable, and the closed-form pairing makes it readable.

The same mixture sum gives the Greeks in closed form—each a q-weighted sum of component Black–Scholes sensitivities, no bumping and no Monte-Carlo noise. With $d_1^i = [\log(F_j^i/K) + \frac{1}{2}(\sigma_j^i)^2]/\sigma_j^i$,

$$\Delta = \frac{1}{S}\sum_i q_j^i F_j^i\, \Phi(d_1^i), \qquad \mathcal{V} = \sum_i q_j^i F_j^i\, \varphi(d_1^i), \qquad \Gamma = \frac{1}{S^2}\sum_i \frac{q_j^i F_j^i\, \varphi(d_1^i)}{\sigma_j^i}, \tag{4}$$

with Δ, Γ in spot and $\mathcal{V}$ per unit total volatility ($\times\sqrt{T_j}$ for the annualised vega); higher sensitivities follow the same way. These are the *sticky-moneyness* Greeks of the fixed marginal, computed with the smile held fixed in moneyness (SSR = 0). The desk hedge delta instead rolls the smile at the calibrated SSR—the SSR-consistent Δ_{SSR} of (26).

3 The dynamic layer: admissible transition kernels

With the marginals fixed, this section defines what may move between them. We set up the Markov kernel on the joint spot–regime state and carve out the *admissible* set—positive, normalised, martingale, marginal-matching—on which absence of static arbitrage is a theorem and existence is Strassen's. Every dynamic choice in the paper is made *within* this set; Section 4 selects the element.

3.1 State and kernel

The dynamic object is a Markov chain on the joint state

$$X_j = (Z_j, f_j, s_j), \qquad Z_j = \log(S_{T_j}/F_j), \quad f_j \in \{1, \ldots, n_{\mathsf{F}}\}, \quad s_j \in \{1, \ldots, n_{\mathsf{S}}\}, \tag{5}$$

where Z_j is log-spot and (f_j, s_j) are a *fast* and a *slow* latent volatility regime—*both carried* across maturities, the genuine two-timescale (Bergomi) memory. The transition over $[T_j, T_{j+1}]$ is a Gaussian-mixture kernel that places the spot–vol leverage in the return *innovation*—a within-step node $l = 1, \ldots, n_{\mathsf{L}}$ correlated with the regime moves—around a *fixed* stationary vol target:

$$\mathcal{K}\big((z,f,s), (\,\mathrm{d}z', f', s')\big) = \sum_{l=1}^{n_{\mathsf{L}}} w_l \ \underbrace{P_{\mathsf{F}}(f' \mid l, f)\, P_{\mathsf{S}}(s' \mid l, s)}_{\text{two carried OU factors}}\, \mathcal{N}\big(\,\mathrm{d}z';\, z + d_{f,s,l},\, V_{f,s,l}\big), \tag{6}$$

with $\mathcal{N}(\,\mathrm{d}z'; m, V)$ the Gaussian measure with mean m and variance V. The *coefficients* on the right—the weights w_l, the transitions $P_{\mathsf{F}}, P_{\mathsf{S}}$, the drift $d_{f,s,l}$ and variance $V_{f,s,l}$—are functions of the regime indices alone; the spot z enters *only* as the additive centre of the Gaussian, so the kernel depends on (z, z') solely through the return $z' - z$. It is therefore **translation-invariant in log-spot**: shifting spot shifts the landing law by the same amount, so the conditional return law—and hence the smile in moneyness—depends on the regime (f, s) alone, not on the spot level. This single property keeps every dynamic readout, the SSR in particular, an exact closed form (Section 6), and makes propagation an exact Gaussian-mixture map (Section 4). The within-step node l carries the instantaneous skew and is renewed each step; the two regimes are carried across steps.

Spot marginal vs. lifted law. The kernel acts on the joint state, so its one-step law is a *lifted* joint $\widehat{\mu}_j(\mathrm{d}z, f, s)$, whereas SANOS supplies only the spot marginal $\mu_j^Z(\mathrm{d}z) = \sum_{f,s} \widehat{\mu}_j(\mathrm{d}z, f, s)$. The first maturity is lifted with the stationary joint regime law $\pi_{f,s}$; thereafter the joint is carried by the propagation. The shorthand $\mu_j \mathcal{K}_j = \mu_{j+1}$ denotes the match of *spot* marginals (Remark 1).

3.2 Admissibility and no static arbitrage

Definition 1 (Admissible kernel)**.** *Given forward-normalised marginals μ_j, μ_{j+1}, a Markov kernel $\mathcal{K}_j$ is* admissible *if*

$$\mathcal{A}_j = \Big\{ \mathcal{K}_j : \ \mathcal{K}_j \geq 0, \ \mathcal{K}_j \mathbf{1} = 1, \ \mathbb{E}[e^{Z_{j+1}} \mid X_j = x] = e^{Z_j} \ \forall x, \ \mu_j \mathcal{K}_j = \mu_{j+1} \Big\}. \tag{7}$$

The conditions are, in order: positivity, normalisation, the (forward-normalised) martingale property, and exact marginal propagation.

Theorem 1 (No static arbitrage)**.** *Let $\{\mu_j\}_{j=0}^M$ be probability measures on $(0, \infty)$ with $\mathbb{E}_{\mu_j}[S] = F_j$ whose forward-normalised laws $\tilde{\mu}_j = \mathrm{Law}(S_{T_j}/F_j)$ are increasing in convex order, $\tilde{\mu}_0 \preceq_{\mathrm{cx}} \tilde{\mu}_1 \preceq_{\mathrm{cx}} \cdots \preceq_{\mathrm{cx}} \tilde{\mu}_M$. If $\mathcal{K}_0, \ldots, \mathcal{K}_{M-1}$ are admissible (each $\mathcal{K}_j \in \mathcal{A}_j$), then the discrete-time process with initial law μ_0 and transitions $\{\mathcal{K}_j\}$ is a martingale after discounting, has one-dimensional marginals exactly $\{\mu_j\}$, and the induced call surface $C_j(K) = D_j \int (s - K)^+ \mu_j(\mathrm{d}s)$ is free of strike (butterfly) and calendar static arbitrage. Moreover $\mathcal{A}_j \neq \emptyset$ for every j.*

A proof sketch is given in Appendix B. The content is that the no-arbitrage *claim attaches to the admissible set*, and that the set is nonempty precisely because SANOS delivers marginals in convex order (Strassen). SANOS's own canonical discrete-local-volatility coupling [1] is one element of $\mathcal{A}_j$—the rigid one the introduction set out to replace; the dynamic targets of Sections 5–6 (SSR, VIX, forward starts) *select a different, controllable element within $\mathcal{A}_j$*—they never replace the set.

Remark 1 (Structural marginal-matching, not fit-then-project)**.** *The translation-invariant kernel of Section 4 cannot match an arbitrary prescribed marginal by itself: a pure convolution adds the same increment law at every spot and cannot reshape the smile. Marginal-matching is restored by a deterministic* leverage overlay *$\sigma_{\mathrm{LV}}(z, T_j)$—the discrete stochastic-local-volatility structure—chosen by Gyöngy's identity so the forward map lands on the SANOS target* by construction *(Section 4.6,* (17)*). There is thus* no fit-then-project*: the first three admissibility conditions are structural in the coefficient map and the fourth (marginal-matching) is the leverage, so calibration fits only the* dynamic *(SSR) knobs while the statics come along for free.*

4 The two-timescale latent-regime kernel

This section constructs the element of $\mathcal{A}_j$ we use: the two-factor Bergomi latent variance, Gauss–Hermite-discretised into a finite regime chain driven by nine structural knobs, with the spot–vol leverage placed in the return innovation. Three ingredients are arranged to be exact: the per-fibre martingale lock, the Gaussian-mixture closure under propagation, and the Gyöngy leverage that lands the propagated marginals on the SANOS targets. Recompression and semigroup interpolation then complete the toolkit.

4.1 Two carried timescales and the Gauss–Hermite discretisation

The latent log-variance is a baseline plus two mean-reverting Ornstein–Uhlenbeck factors—a *fast* factor X^{F} (rate κ_{F}) and a *slow* factor X^{S} (rate κ_{S})—so that $x_j = \bar{\gamma} + X_j^{\mathsf{F}} + X_j^{\mathsf{S}}$, the continuous two-factor Bergomi model. Conditional on the latent vol the one-step log-return is Gaussian, but its mean and variance are *functions of the continuous vol state*: the kernel is then a continuous, variable-coefficient mixture of Gaussians—an integral over the latent factors, *not* a finite Gaussian mixture, and not closed form. The remedy is quadrature. An n-point **Gauss–Hermite** rule represents $\mathcal{N}(0,1)$ by n nodes ζ_l and weights w_l reproducing the Gaussian's moments through order $2n-1$,

$$\mathbb{E}_{\mathcal{N}(0,1)}[g] = \int g(\zeta)\,\frac{e^{-\zeta^2/2}}{\sqrt{2\pi}}\,\mathrm{d}\zeta \;\approx\; \sum_{l=1}^{n} w_l\, g(\zeta_l) \qquad (\text{exact for } \deg g \le 2n-1), \tag{8}$$

the moment-optimal finite stand-in for a Gaussian. Replacing each continuous latent quantity by its n GH nodes turns the integral into a finite sum—each node a *vol scenario*—so the kernel becomes a finite Gaussian mixture over a quadrature grid. *Both* vol factors are carried:

- **Fast factor** F—carried regime, node index $f = 1,\ldots,n_{\mathsf{F}}$, mean-reversion κ_{F} (the high, decaying short end of the SSR);
- **Slow factor** S—carried regime, node index $s = 1,\ldots,n_{\mathsf{S}}$, mean-reversion κ_{S} (the sustained long-end floor);
- **Within-step return** L—node index $l = 1,\ldots,n_{\mathsf{L}}$, *renewed* each step, carrying the instantaneous skew and the innovation leverage.

The nodes $\{\zeta_f^{\mathsf{F}}\}, \{\zeta_s^{\mathsf{S}}\}, \{\zeta_l^{\mathsf{L}}\}$ and base log-weights $\eta^{\mathsf{F}}, \eta^{\mathsf{S}}$ (within-step weights $w_l = \mathrm{softmax}(\eta^{\mathsf{L}})$) are *fixed* quadrature constants, not parameters (fitting the base weights as an expressive residual is a design alternative we do not use); the data fit only the nine structural knobs of the coefficient map below. The vol-of-vols $\nu_{\mathsf{F}}, \nu_{\mathsf{S}}, \nu_{\mathsf{L}}$ scale the nodes, the leverages $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$ couple the regime moves to the realised return node l (never to z), and λ_{skew} tilts the within-step return. The node counts are a deliberate *modelling* choice, not an accuracy dial. The within-step factor (small ν_{L}) is converged at $n_{\mathsf{L}} = 5$, but the carried factors converge *slowly*: $V \propto e^{\nu\zeta}$ is lognormal in the node, and Gauss–Hermite, being polynomial-exact, resolves $\mathbb{E}[V]$ only gradually for the larger $\nu_{\mathsf{F}}, \nu_{\mathsf{S}}$. A sweep shows the ATM-vol level still drifting at $n_{\mathsf{F}} = 9$, $n_{\mathsf{S}} = 7$. We therefore keep a coarse $n_{\mathsf{F}} = 5$, $n_{\mathsf{S}} = 3$ and let calibration absorb the discretisation: $\bar{\gamma}$ and the leverages are fit *at* the grid, so the model hits its ATM and SSR targets at any node count, and refining merely re-shifts $\boldsymbol{\theta}$. The continuous limit $n_{\mathsf{F}}, n_{\mathsf{S}} \to \infty$ is thus a structural relationship, not a practical convergence claim at these counts; the counts trade resolution against parameter identifiability (Appendix D) and cost.

4.2 The coefficient map

The component coefficients are *determined* by nine structural knobs $\boldsymbol{\theta} = (\bar{\gamma}, \nu_{\mathsf{F}}, \nu_{\mathsf{S}}, \nu_{\mathsf{L}}, \lambda_{\text{skew}}, \lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}, \kappa_{\mathsf{F}}, \kappa_{\mathsf{S}})$. The per-node variance and the within-step skew tilt are

$$V_{f,s,l} = \exp\!\big(\bar{\gamma} + \nu_{\mathsf{F}}\,\zeta_f^{\mathsf{F}} + \nu_{\mathsf{S}}\,\zeta_s^{\mathsf{S}} + \nu_{\mathsf{L}}\,\zeta_l^{\mathsf{L}}\big)\Delta_j, \qquad (\text{two-factor variance}), \tag{9}$$

$$\ell_{f,s,l} = \lambda_{\text{skew}}\,\zeta_l^{\mathsf{L}}\,\sqrt{V_{f,s,l}}, \qquad (\text{instantaneous mean-tilt / skew}), \tag{10}$$

and the two carried factors evolve by leverage-coupled mean-reverting transitions whose tilt is driven by the realised return node l, *not* by log-spot:

$$P_{\mathsf{F}}(f' \mid l, f) = \mathrm{softmax}_{f'}\big(\eta^{\mathsf{F}}_{f'} + \kappa_{\mathsf{F}}\, \zeta^{\mathsf{F}}_{f}\zeta^{\mathsf{F}}_{f'} + \lambda_{\mathsf{F}}\, \zeta^{\mathsf{L}}_{l}\, \zeta^{\mathsf{F}}_{f'}\big), \tag{11}$$

$$P_{\mathsf{S}}(s' \mid l, s) = \mathrm{softmax}_{s'}\big(\eta^{\mathsf{S}}_{s'} + \kappa_{\mathsf{S}}\, \zeta^{\mathsf{S}}_{s}\zeta^{\mathsf{S}}_{s'} + \lambda_{\mathsf{S}}\, \zeta^{\mathsf{L}}_{l}\, \zeta^{\mathsf{S}}_{s'}\big). \tag{12}$$

Here η pulls to the centre (mean reversion), κ to the current regime (persistence / timescale), and $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$ are the *innovation* leverages: a down move (ζ^{L}_{l} large via $\lambda_{\text{skew}} < 0$) tilts the regimes to higher vol. Because the leverage multiplies the innovation around a *fixed* stationary target, the equilibrium vol is *not* a deterministic function of spot—it is genuine stochastic volatility, and the kernel stays translation-invariant in z.

4.3 Exact per-fibre martingale normalisation

The within-step drift is re-locked to the forward on *every* fibre—the kernel's conditional law at a fixed state (z, f, s). With raw mean $\widetilde{m}_{f,s,l} = -\frac{1}{2}V_{f,s,l} + \ell_{f,s,l}$, subtract the per-regime constant

$$A_{f,s} = \log \sum_{l} w_l \, \exp\big(\widetilde{m}_{f,s,l} + \tfrac{1}{2}V_{f,s,l}\big) = \log \sum_{l} w_l \, e^{\ell_{f,s,l}}, \qquad d_{f,s,l} = \widetilde{m}_{f,s,l} - A_{f,s}, \tag{13}$$

so that $\mathbb{E}[e^{Z_{j+1}} \mid z, f, s] = e^{z}$ exactly. The shift is a single per-fibre constant, *independent of* z (the within-step weights w_l are the fixed Gauss–Hermite weights)—stable, differentiable, and it preserves the relative branch tilts that carry the leverage and the SSR. It replaces any penalty-only enforcement and the fragile "adjust one component mean" construction, which fails when the required numerator turns non-positive. The kernel is now complete: a finite Gaussian-mixture martingale on (z, f, s).

4.4 Spot-translation invariance and the SSR dial

With every coefficient in (6) independent of z (Section 3), the conditional smile $\sigma(T; f, s)$ depends only on the regime pair, and the skew-stickiness ratio is generated by the *innovation* leverage $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$ (the return$\leftrightarrow$regime correlation), not by any z-coupling: $\lambda_{\mathsf{F}} = \lambda_{\mathsf{S}} = 0 \Rightarrow$ no spot–vol comovement $\Rightarrow$ SSR $= 0$ (sticky-moneyness); the dials and their term-structure shaping are developed in Section 6. The SSR is thus not a spot-derivative of a state-dependent smile but the closed-form return $\times$ regime covariance of Section 6. The placement of the leverage is a design decision with teeth. A Monte-Carlo cross-check showed that coupling the regimes to *spot* instead makes the equilibrium vol a deterministic function of spot (local volatility), which pins SSR $\to 2$ at long maturities and cannot be re-tuned. Placing the leverage in the innovation breaks that floor while keeping the readouts exact.

4.5 Exact Gaussian-mixture closure

The point of translation invariance is that one propagation step is an *exact Gaussian convolution*: the finite mixture of Section 4 stays a finite mixture, so every downstream readout is a closed-form sum rather than a quadrature. Conditional on the regime, the kernel adds an increment $\Delta z \sim \mathcal{N}(d_{f,s,l}, V_{f,s,l})$ to the log-price (Section 4.4), and Gaussians convolve in closed form—means and variances add:

$$\mathcal{N}(\,\cdot\,; m^{f,s}_{c}, v^{f,s}_{c}) \;*\; \mathcal{N}(\,\cdot\,; d_{f,s,l}, V_{f,s,l}) \;=\; \mathcal{N}\big(\,\cdot\,; m^{f,s}_{c} + d_{f,s,l},\; v^{f,s}_{c} + V_{f,s,l}\big). \tag{14}$$

So a law carried, per regime (f,s), as a Gaussian mixture in log-price $\sum_c \omega_c^{f,s}\,\mathcal{N}(m_c^{f,s}, v_c^{f,s})$ propagates to another finite mixture,

$$\rho_{j+1}(z', f', s') = \sum_{f,s,\,c} \omega_c^{f,s} \sum_l w_l\, P_{\mathsf{F}}(f'\,|\,l, f)\, P_{\mathsf{S}}(s'\,|\,l, s)\,\mathcal{N}\big(z';\, m_c^{f,s} + d_{f,s,l},\, v_c^{f,s} + V_{f,s,l}\big), \tag{15}$$

each new component the convolved Gaussian of (14), reweighted by the within-step weight w_l and the regime transitions: GM $*$ GM = GM, exactly—no quadrature, PDE, or Monte Carlo anywhere in the chain. (A *state-dependent* kernel—z-dependent weights $w_l(z)$ or variances—breaks (14): propagation relapses to a continuous integral mixture, forcing a first-order collocation that a Monte-Carlo cross-check showed biases the multi-step SSR.) The lone cost is that (15) multiplies the component count by $n_{\mathsf{L}} n_{\mathsf{F}} n_{\mathsf{S}}$ each step; the recompression of §4.7 below holds it to a fixed budget by a moment-preserving, martingale-relocked reduction. Convex-order / calendar no-arbitrage rests on the martingale kernel, not on this exact finite closure.

4.6 The discrete SLV fusion: leverage carries the statics

Because the kernel is translation-invariant, $\mu_j \mathcal{K}_j$ is a pure convolution—it cannot bend its own (stochastic-vol) marginals onto a prescribed local-volatility surface. Matching the SANOS marginals therefore needs one structural ingredient, a deterministic leverage—exactly the stochastic-local-volatility decomposition. Split the increment into a local scale and a unit stochastic-vol factor,

$$\Delta z \;\sim\; \bar{d}_{f,s} + \underbrace{\sigma_{\mathrm{LV}}(z, T_j)\sqrt{\Delta_j}}_{\text{statics (SANOS)}} \cdot \underbrace{\sqrt{\nu_{f,s}}}_{\text{SSR (kernel)}}\;\xi, \qquad \nu_{f,s} = \frac{\bar{V}_{f,s}}{\mathbb{E}_\pi[\bar{V}]}, \quad \bar{V}_{f,s} = \mathbb{E}_l[V_{f,s,l}] + \mathrm{Var}_l(d_{f,s,l}), \tag{16}$$

where ξ is a standard (unit-variance) innovation, $\bar{d}_{f,s}$ the per-fibre martingale-lock drift (Section 4.4), and $\bar{V}_{f,s}$ the kernel's *full* one-step increment variance by the law of total variance—the within-node $\mathbb{E}_l[V_{f,s,l}]$ plus the between-node mean spread $\mathrm{Var}_l(d_{f,s,l})$ the skew tilt λ_{skew} creates (using $\mathbb{E}_l[V]$ alone mis-scales the leverage by $\sim 3\times$ at the calibrated λ_{skew}). The unit factor $\nu_{f,s}$ ($\mathbb{E}_\pi[\nu] = 1$, with italic regime indices—not the sans-labelled vol-of-vol amplitudes $\nu_{\mathsf{F}}, \nu_{\mathsf{S}}, \nu_{\mathsf{L}}$) keeps the kernel's fluctuation—the regime comovement, i.e. the SSR—and drops its level; σ_{LV} sets the level from the market. Marginal-matching is then Gyöngy's identity [6] in discrete form,

$$\sigma^2_{\mathrm{LV}}(z, T_j) = \frac{\sigma^2_{\mathrm{Dupire}}(z, T_j)}{\mathbb{E}[\nu \mid z]}, \tag{17}$$

with σ_{Dupire} from the SANOS marginals (closed-form Gaussian-mixture call + Breeden–Litzenberger density, calendar finite-difference) and $\mathbb{E}[\nu \mid z]$ the *spot-conditional* average of the unit factor (distinct from the stationary $\mathbb{E}_\pi[\nu] = 1$; near 1, tilted only by the spot–vol correlation)—(17) being the exact discrete Gyöngy projection regardless. The decoupling is then literal: SANOS sets the statics and, through the leverage, the local-vol roll; the kernel carries the SV response (the bulk of the SSR level). In parameter terms the decoupling *is* the split of the fused per-step variance into a market level and a unit-mean kernel shape,

$$\mathrm{Var}\big(\Delta z \mid z, f, s\big) = \underbrace{\sigma^2_{\mathrm{LV}}(z, T_j)\,\Delta_j}_{\text{level (market)}} \cdot \underbrace{\nu_{f,s}}_{\text{shape (kernel)},\ \mathbb{E}_\pi[\nu]=1}, \tag{18}$$

whose π-average is exactly $\sigma^2_{\mathrm{LV}}\Delta_j$: the *level* is set by the leverage, and the kernel reaches the fused law *only* through the unit-mean ratio $\nu_{f,s} = \bar{V}_{f,s}/\mathbb{E}_\pi[\bar{V}]$. The kernel's own variance level is carried by $\bar{\gamma}$, which multiplies every $V_{f,s,l}$ by $e^{\bar{\gamma}}$; this nearly divides out of ν and is reset by $\sigma_{\rm LV}$, leaving $\bar{\gamma}$ nearly inert (the $\mathrm{Var}_l(d)$ term breaks exact $e^{\bar{\gamma}}$ homogeneity). What survives in ν is the regime-*relative* structure—the $\nu_{\mathsf{F}}\zeta^{\mathsf{F}}_f, \nu_{\mathsf{S}}\zeta^{\mathsf{S}}_s$ spread of $\bar{V}$ and, through the transitions $P_{\mathsf{F}}, P_{\mathsf{S}}$, the leverages $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$ and rates $\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$—which is the SSR. (Numerically, a unit shift in $\bar{\gamma}$ moves ν by $< 0.2\%$, the same shift in ν_{F} by tens of percent.) In a *spot-uncorrelated* state $\mathbb{E}[\nu \mid z] \approx 1$, so the leverage barely rescales the kernel ($\sigma_{\rm LV} \approx \sigma_{\rm Dupire}$), the smile is matched essentially for free, and the SSR is the lone free dynamic target.

Because $\sigma_{\rm LV}(z)$ is a smooth, deterministic statics object, collocating it at the component means affects only the level, not the SSR mechanism, which stays translation-invariant and exact. For *marginal matching*—each step re-seeded from the target law, with regimes at their spot-uncorrelated stationary spread—the residual is then discretisation rather than bias: the discrete Dupire under-reads the local variance at the narrowest (short-T) marginals ($\sim$50–85 bp), and this tightens with a central-difference Dupire, finer grids, and a tail-clamped leverage. Under multi-step *concatenation from a fixed spot* (forward densities and hedging, Section 4.7) the shortcut is *not* free. The negative leverage tilt correlates high-ν regimes with low z, so the spot-conditional $\mathbb{E}[\nu \mid z{=}0]$ falls measurably below 1 (empirically ≈ 0.7 on SPX), and approximating $\sigma_{\rm LV} \approx \sigma_{\rm Dupire}$ then *under-reads* the accumulated ATM level by of order 10% in vol over a multi-month chain. There the exact projection (17) is retained: $\mathbb{E}[\nu \mid z]$ is negligible for reproducing marginals, but a genuine path-conditional rescaling for absolute-level outputs such as forward implied vol and hedge P&L.

4.7 Recompression: keeping the budget flat under concatenation

Recompression is a *multi-step* device: a *single* kernel application leaves a manageable $n_{\mathsf{L}}n_{\mathsf{F}}n_{\mathsf{S}}$-component mixture and needs none. It is *concatenation*—composing $\mathcal{K}_j \circ \mathcal{K}_{j+1} \circ \cdots$ over several steps for forward densities, exotics, or interpolation—that compounds the count: each composed step multiplies the per-fibre spot mixture by $n_{\mathsf{L}}n_{\mathsf{F}}n_{\mathsf{S}}$ (each target fibre aggregates the $n_{\mathsf{F}}n_{\mathsf{S}}$ source regimes and the n_{L} within-step nodes), so left alone the count is $(n_{\mathsf{L}}n_{\mathsf{F}}n_{\mathsf{S}})^J$ after J steps. Each composed step therefore ends with a recompression that holds the budget flat—it preserves the *propagated* marginal with fewer components; the match to the SANOS *target* is the separate Gyöngy step (Remark 1), not part of recompression. The default is a deterministic *moment-preserving merge* (per fibre: sort by mean, split into $n_{\rm keep}$ equal-weight chunks, match each chunk's first three moments), and one may equivalently refit a compact martingale kernel $\widetilde{\mathcal{K}}_j$ reproducing the propagated marginal at the digital level,

$$\min_{\widetilde{\mathcal{K}}_j} \; d_{\rm CDF}\big(\rho_j\widetilde{\mathcal{K}}_j, \rho_j\mathcal{K}_j\big) \quad \text{s.t.} \quad \widetilde{\mathcal{K}}_j \geq 0,\; \textstyle\sum_l \widetilde{w}_l\, e^{\widetilde{d}_l + \frac{1}{2}\widetilde{\sigma}_l^2} = 1 \text{ (per fibre)}. \tag{19}$$

Because $\widetilde{\mathcal{K}}_j$ is a martingale kernel, $\rho_j\widetilde{\mathcal{K}}_j \succeq_{\rm cx} \rho_j$ automatically (Strassen), so convex order—hence calendar no-arbitrage—holds by construction and the convex-order inequalities need not be imposed. The martingale is re-locked by (13) after the merge (which preserves the moments of z but not the forward $\mathbb{E}[e^z]$); a constrained weight-LP (fixed reduced locations, weights fitted to the propagated CDF subject to exact forward and convex order, reporting the residual) is available as a refinement where the merge is too coarse. Recompression is done every composed step, per-fibre (never merging away the carried regimes), deterministically, with the residual reported.

Recompression touches only the component *budget*; the marginal-match is restored *structurally*, by re-applying the Gyöngy leverage (17) to the recompressed mixture. The leverage is a local, deterministic rescaling read from the recompressed mixture's own Dupire surface against the SANOS target, so it absorbs the moment-merge error in the same operation that matches the kernel's intrinsic marginal: the per-step pipeline is *propagate* (dynamics) $\to$ *recompress* (budget + re-lock)

$\to$ *leverage* (Gyöngy) $\to$ re-lock, with *no projection at any step.* Convex order survives because the match is to the convex-ordered SANOS target, and the leverage edits only the local *scale*, leaving the regime dynamics—the SSR—exact.

4.8 Intermediate maturities: OU-semigroup interpolation

Because the carried factor is OU, the generator is time-homogeneous and the transition operators form a semigroup. Writing the latent log-variance as $x_t = g_0(t) + X_t^{\mathsf{F}} + X_t^{\mathsf{S}}$ (a time-inhomogeneous forward-variance level g_0 plus a time-homogeneous generator), one interpolates only the one-dimensional curve g_0 and constructs intermediate kernels by evaluation. At the midpoint the kernel square root is the half-time semigroup element $\mathcal{K}_{[T_1,T_2]} = P_{\Delta T/2}$, so $\mathcal{K} \circ \mathcal{K} = P_{\Delta T} = \mathcal{K}_{[T_1,T_3]}$ exactly, with no optimisation. The carried vol state is what makes this work: the spot-only kernel (marginalising the vol state at the intermediate step) does not compose, which is why a memory-less kernel requires a matching optimisation. Calendar no-arbitrage on the interpolated marginal is automatic when g_0 gives non-decreasing total variance.

5 The fitting step: digitals as a price metric

We fit *digitals* (CDFs), not prices. A digital is the strike-derivative of a call, $-\partial_K C = 1 - G(K)$—one derivative above the price, one below the density—and the bound that justifies the choice lives in the *spot/strike* variable s (with G the law of S_T):

$$\begin{aligned}\sup_K |C_{\boldsymbol\theta} - C_\star| \le\ & \|C_{\boldsymbol\theta} - C_\star\|_{\dot W^{1,1}} = \int_0^\infty |G_{\boldsymbol\theta}(s) - G_\star(s)|\,\mathrm{d}s \\ =\ & W_1(\rho_{\boldsymbol\theta}, \rho_\star) = \int_0^1 |G_{\boldsymbol\theta}^{-1}(u) - G_\star^{-1}(u)|\,\mathrm{d}u.\end{aligned} \tag{20}$$

The middle equality is what makes the digital fit a *price* metric: the L^1 norm of $\partial_K(C_{\boldsymbol\theta} - C_\star)$ is the homogeneous Sobolev $\dot W^{1,1}$ seminorm of the call-price difference, equal to its total variation and—since $-\partial_K C$ is the digital—to the *aggregate mispricing of all digitals.* The gap $G_\star - G_{\boldsymbol\theta}$ that the band-loss (21) drives into the bid–ask band is exactly the integrand of this W_1, so fitting digitals strike by strike minimises the price-space Wasserstein distance. The digital is the natural target because it sits one strike-derivative below the noisy density and one above the coarse price.

Remark 2 (Coordinate caveat)**.** *The call payoff $(s-K)^+$ is 1-Lipschitz in spot s, which is why the sup price gap is bounded by W_1 in spot. In log-coordinate $z = \log s$ the integral acquires the Jacobian weight, $\int |G_{\boldsymbol\theta} - G_\star| e^z\,\mathrm{d}z$, so an unweighted log-space CDF loss does not by itself control call prices. Accordingly, any log-space, exponential-growth discrepancy used adversarially below is a financial pricing discrepancy / integral probability metric, not literally Wasserstein-*1*; we reserve "Wasserstein" for the spot-coordinate bound* (20)*.*

Objective. Collect the generator parameters $\boldsymbol\theta$. The calibration stacks digital band-losses over the SPX (and optional forward-start) instruments, plus the dynamic targets:

$$\min_{\boldsymbol\theta} \sum_{i\in\mathrm{SPX}} w_i^{\mathrm{spr}}\,\mathrm{dist}\big(1 - G_{\boldsymbol\theta}^S(K_i), [\underline{S}_i, \overline{S}_i]\big)^2 + \beta_{\mathrm{SSR}} \sum_T \big(\mathrm{SSR}_T(\boldsymbol\theta) - \mathrm{SSR}_T^\star\big)^2 + \beta_{\mathrm{vov}}\,\mathcal{R}_{\mathrm{vov}}(\boldsymbol\theta). \tag{21}$$

Here $G_{\boldsymbol\theta}^S$ is the CDF of the model's terminal law under the kernel $\boldsymbol\theta$—the *Gyöngy-fused* propagated marginal, whose leverage (17) carries the SANOS statics. Writing it as the mixture

$\sum_i q^i_{\boldsymbol{\theta}} \mathcal{N}(\,\cdot\,; \mu^i_{\boldsymbol{\theta}}, (\sigma^i_{\boldsymbol{\theta}})^2)$ in log-moneyness, with component forwards $F^i_{\boldsymbol{\theta}} = Fe^{\mu^i_{\boldsymbol{\theta}} + \frac{1}{2}(\sigma^i_{\boldsymbol{\theta}})^2}$, the band-loss uses the model digital

$$1 - G^S_{\boldsymbol{\theta}}(K_i) = \Pr_{\boldsymbol{\theta}}(S > K_i) = \sum_i q^i_{\boldsymbol{\theta}}\, \Phi(d^i_2), \qquad d^i_2 = \frac{\log(F^i_{\boldsymbol{\theta}}/K_i) - \frac{1}{2}(\sigma^i_{\boldsymbol{\theta}})^2}{\sigma^i_{\boldsymbol{\theta}}},$$

the Breeden–Litzenberger survival $-\partial_K C_{\boldsymbol{\theta}}$ of the mixture call (3), fit to the market digital bid–ask band $[\underline{S}_i, \overline{S}_i]$. The w^{spr}_i are inverse-spread weights and $\beta_{\mathrm{SSR}}, \beta_{\mathrm{vov}}$ the soft-target weights; $\mathrm{SSR}_T(\boldsymbol{\theta})$ is the model SSR (22) (in closed form, (23)) and $\mathrm{SSR}^\star_T$ its target; $\mathcal{R}_{\mathrm{vov}}(\boldsymbol{\theta})$ is the vol-of-vol readout residual (30); both *dynamic* targets—the SSR and the vol-of-vol readout—are developed in Section 6. The minimisation is subject to the admissibility constraints (7). The martingale property is one of them, enforced *exactly* rather than softly: a single-marginal mixture calibration would add a forward-matching cost $\lambda\big(\sum_k \pi_k e^{m_k + q_k/2} - 1\big)^2$ to the loss, whereas here the per-fibre lock (13) drives the conditional forward error to zero at every state and every $\boldsymbol{\theta}$, so the analogous term is identically zero and is dropped. Only the genuinely free targets—marginal fit, SSR, and vol-of-vol—remain in (21).

Two regimes for the dynamic targets. Where forward-start and VIX digitals are *liquid*, fit those prices directly through the band-loss and let the SSR *emerge*, validating it against the empirical value as a diagnostic—the principled, risk-neutral route. Even then the emergent SSR is not guaranteed: the Quintic OU model fits the full SPX *and* VIX smiles jointly and still finds its SSR significantly below the market's, restoring it with exactly such a band penalty [28]. Where the dynamic instruments are thin (most underlyings), the β_{SSR} penalty instead acts as an interpretable *steer*, its target a realized-SSR estimate or a desk view—a real-world ($\mathbb{P}$) quantity, not the risk-neutral SSR, so it neglects the volatility risk premium. The penalty form is robust either way; an adversarial form (a min–max over an exponential-growth critic for the worst-mispriced digital) is an optional alternative when forward-start/VIX digitals are rich.

Algorithm 1 orchestrates the full daily calibration, with the SANOS LP (Algorithm 3) as its static first step. The kernel fit is the one *non-convex* part—in contrast to the convex SANOS LP—and the structural marginal-matching (Remark 1: the leverage (17) carries the statics, then moment-preserving recompression re-locks the martingale) guarantees the returned model is exactly arbitrage-free wherever the optimiser lands.

What lands on the desk. The kernel is an internal engine behind a standard vol-surface API; the desk receives the deliverables of Table 1, all closed-form and all functionals of the *same* calibrated object the algorithm returns. Continuously monitored barriers are the one caveat—they need sub-stepping or a bridge correction—while the other exotics are direct kernel functionals.

Table 1: What lands on the desk: every output is a functional of one calibrated object (SANOS marginals + the transition kernel).

Deliverable	Source
SPX vanilla surface + Greeks	spot marginals (drop-in)
Forward-vol surface $\widehat{\Sigma}^{\mathrm{fwd}}(z,k)$	transition kernel
VIX-identified vol-of-vol	kernel (VIX-informed, Sec. 6)
Exotics: forward-starts, cliquets, autocallables, barriers	propagate through the kernel
SSR-consistent Δ, vanna	kernel dynamics (Sec. 6)
De-event smile, event-jump density, event-conditional hedge	clean-flank bridge (Sec. 7)

Algorithm 1 SANOS-Evolve calibration (one pricing date).

Require: per-maturity option chains (calls and puts) $\{(T_j, K, \mathrm{bid/ask})\}_{j=1}^{M}$ and a discount curve (the static layer); *and*—since the chains pin the marginals but not the dynamics—a dynamic anchor for the SSR/vol-of-vol: forward-starts or a vol-index (VIX for SPX) where liquid, else a realised estimate from the underlying's history. The forward curve (put–call parity) and forward-variance level (variance swap) are derived from the chains

Ensure: marginals $\{\mu_j\}$ and generator $\boldsymbol{\theta} = (\bar{\gamma}, \nu_{\mathsf{F}}, \nu_{\mathsf{S}}, \nu_{\mathsf{L}}, \lambda_{\mathrm{skew}}, \lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}, \kappa_{\mathsf{F}}, \kappa_{\mathsf{S}})$ defining admissible kernels $\{\mathcal{K}_j\}$; the desk outputs of Table 1

1: **Static layer.** For each T_j, fit μ_j by the SANOS LP (Algorithm 3); verify increasing convex order $\mu_j \preceq_{\mathrm{cx}} \mu_{j+1}$ (the precondition of Theorem 1); repair the marginals if violated.

2: **Readout.** Extract the vol-of-vol level $(v_\infty, \xi_{\mathrm{VIX}})$ from the forward-variance readout—VIX futures and call-spreads for SPX, else the strip's variance-swap level with smile curvature and realised vol-of-vol (Table 2); the two timescales $\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$ are *both* fit, their fast/slow split resolved by the VIX vol-of-vol *term structure* (the damping $D(\tau)$ at two horizons) on top of the SSR slope (Appendix D).

3: **Regime scaffold.** Fix Gauss–Hermite nodes $\{\zeta_f^{\mathsf{F}}\}, \{\zeta_s^{\mathsf{S}}\}, \{\zeta_l^{\mathsf{L}}\}$ and base log-weights η; initialise $\boldsymbol{\theta}$ from a global vol-quantile split and the readout.

4: **for** $j = 1, \ldots, M-1$ (warm-started across maturities) **do**

5:     **repeat**

6:         **Kernel fit.** Minimise the objective (21) over $\boldsymbol{\theta}$: digital band-loss to μ_{j+1}, plus β_{SSR} (SSR term structure) and β_{vov} (vol-of-vol readout), subject to admissibility (7) (positivity, normalisation, exact martingale lock). $\triangleright$ non-convex

7:         **Structural match + recompress.** The leverage (17) lands the forward map on μ_{j+1} by construction; moment-preserving recompression (Sec. 4.7) holds the component budget and re-locks the martingale. Record the residual $d(\mu_j \mathcal{K}_j, \mu_{j+1})$.

8:     **until** marginal, SSR, and vol-of-vol residuals are within their bands

9: **end for**

10: **return** $\{\mu_j\}$, $\boldsymbol{\theta}$, $\{\mathcal{K}_j\}$; price vanillas/exotics, Greeks, SSR-deltas, and de-event smiles in closed form (Table 1).

6 Smile dynamics: the SSR and the forward-variance readout

This section develops the two *dynamic* targets of the calibration objective (21): the skew-stickiness ratio $\mathrm{SSR}_T(\boldsymbol{\theta})$ (the regime comovement, in closed form below) and the forward-variance / vol-of-vol readout $\mathcal{R}_{\mathrm{vov}}(\boldsymbol{\theta})$—each a closed-form function of the generator $\boldsymbol{\theta}$. The desk-facing ATM features and the SSR-consistent hedge follow from the same machinery.

6.1 The skew-stickiness ratio

With $k = \log(K/F)$, ATM skew $\mathcal{S}_T = \partial_k \widehat{\sigma}_T(k)|_{k=0}$, the SSR of Bergomi [22] is

$$\mathrm{SSR}_T = \frac{1}{\mathcal{S}_T} \frac{\mathrm{d}\widehat{\sigma}_{\mathrm{ATM},T}}{\mathrm{d}\log S}. \tag{22}$$

In this convention SSR = 1 is *sticky-strike* (fixed-strike vols pinned; the ATM vol slides along the skew) and SSR = 0 is *sticky-delta* (the smile rides with spot; ATM vol unchanged). The empirical SPX term structure of Section 1 sits at or beyond sticky-strike, never near sticky-delta. We fix this convention throughout.

6.2 The SSR is a closed-form realised covariance

A single marginal pins $\mathrm{Law}(S_T)$ but not how the smile moves; that information lives in the kernel. Because the kernel is translation-invariant in z (Section 4.4), the conditional smile $\sigma(T; f, s)$ depends only on the regime pair, so the empirical SSR—the regression of ATM-vol changes on returns—is *not* a spot-derivative but a finite closed-form sum over the regime transitions. With $\pi_{f,s}$ the stationary law of the carried chain, r the one-step return with node values $d_{f,s,l}$, and $\mathrm{skew}(T) = \sum_{f,s} \pi_{f,s}\, \mathcal{S}_T(f, s)$ ($\mathcal{S}_T(f, s) = \partial_k \sigma(T; f, s)|_{k=0}$ the regime-conditional ATM skew),

$$\begin{aligned} \mathrm{SSR}(T) &= \frac{\mathrm{Cov}\left(\Delta\sigma_{\mathrm{ATM}}(T),\, r\right)}{\mathrm{Var}(r)\,\mathrm{skew}(T)}, \\ \mathrm{Cov} &= \sum_{f,s} \pi_{f,s} \sum_{l} w_l\, d_{f,s,l} \sum_{f',s'} P_{\mathsf{F}}(f'|l, f) P_{\mathsf{S}}(s'|l, s) \left[\sigma(T; f', s') - \sigma(T; f, s)\right], \\ \mathrm{Var}(r) &= \sum_{f,s} \pi_{f,s} \Big[\sum_{l} w_l (d^2_{f,s,l} + V_{f,s,l}) - \big(\sum_{l} w_l d_{f,s,l} \big)^2 \Big], \end{aligned} \tag{23}$$

with $\sigma(T; f, s)$ the ATM vol of the (T/Δ_j)-step propagation started in regime (f, s) ($\mathrm{Var}(r)$ the π-averaged conditional return variance; the between-regime mean term $\mathrm{Var}_\pi(\sum_l w_l d_{f,s,l}) \sim 0.2\%$ is higher-order and dropped). The covariance *is* the leverage: a down-return node l tilts $P_{\mathsf{F}}, P_{\mathsf{S}}$ toward higher vol (via $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$), so $\Delta\sigma$ correlates with r. It is a finite sum over (l, f, s, f', s')—no spot-bump, no Monte Carlo, no measure-transport approximation—and is Monte-Carlo-confirmed: the sum sits at the centre of a direct one-step regression, $|\mathrm{diff}| \approx 0.005$ at 1m/3m/6m/1y (Section 8.1). Once the (17) leverage is in, the realised SSR picks up a second, additive piece: a *local-vol roll* $\partial_z \hat{\sigma}$ (the spot-shift derivative $\mathrm{d}\hat{\sigma}_{\mathrm{ATM}}/\,\mathrm{d}\log S$, $z = \log(S/F)$) set by the forward-variance curve, which is statics owned by SANOS. This sits on top of the stochastic-vol response, which carries the bulk of the level. Both pieces are closed form, and the fused readout is likewise Monte-Carlo-validated.

6.3 The SSR dials

The innovation leverages $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$ carry the comovement (Section 4.4): $\lambda_{\mathsf{F}} = \lambda_{\mathsf{S}} = 0 \Rightarrow \mathrm{SSR} = 0$, and raising them lifts it through the sticky-strike value $\mathrm{SSR} = 1$ toward the short-maturity diffusive limit near 2, so the empirical SPX range $\mathrm{SSR} \in [1, 2]$ is a leverage-magnitude statement. Acting through the two timescales with $\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$, they shape the SSR *term structure*—level *and* slope: the fast κ_{F} gives the high, decaying short end, the slow κ_{S} the sustained long-end floor that one factor cannot. That is what Section 5 fits to SPX. (The SSR is regime- and vol-of-vol-dependent rather than fixed by the variance-curve slope; a rising forward-variance curve lifts its *level* relative to a flat one [25]—a leading-order, qualitative effect, not a slope law.) Where forward-start prices are liquid, the same (λ, κ) are pinned by fitting those prices directly. The kernel prices forward-starting calls in closed form, $C^{\mathrm{FS}}_{T_1,T_2}(k) = D_{T_2} \int \rho_{T_1}(\,\mathrm{d}x) \int \mathcal{K}_{T_1,T_2}(x,\, \mathrm{d}y)(e^{y-x} - k)^+$, and its strike-derivative is the forward-return survival function—the clean traded proxy for the SSR—so the SSR *emerges* as a diagnostic. How much the SSR can move at *fixed* smile is bounded: this is the decoupling band of Section 8.1, the soft direction of Appendix D made quantitative.

6.4 Closed-form ATM features and the SSR

Because the conditional forward-return law is a Gaussian (log-normal) mixture, the ATM implied-vol Taylor coefficients—level $\hat{\sigma}(0)$, skew $\hat{\sigma}'(0)$, curvature $\hat{\sigma}''(0)$—are closed form *with no Black–Scholes inversion*, by two routes.

Exact: implicit differentiation. The implied vol solves $C_{\rm mix}(m) = C_{\rm BS}(m, \hat\sigma(m))$ in log-moneyness $m = \log(F/K)$ (so $\partial_m = -\partial_k$ for the §6 skew variable k; sign in App. E). The level is a single probit of the closed-form ATM mixture price, and the higher coefficients follow by implicitly differentiating that identity (derived in full in Appendix E), working in the *total* volatility $u = \hat\sigma\sqrt{T_j}$ (the variable the Black greeks below are taken in),

$$u'(0) = \frac{\partial_m C_{\rm mix} - \partial_m C_{\rm BS}}{\mathcal{V}}, \qquad u''(0) = \frac{\partial_m^2 C_{\rm mix} - \partial_m^2 C_{\rm BS} - 2\,\mathcal{V}_x\,u'(0) - \mathcal{V}_u\,u'(0)^2}{\mathcal{V}},$$
$$\hat\sigma^{(k)}(0) = \frac{u^{(k)}(0)}{\sqrt{T_j}}, \tag{24}$$

with the *total-vol* Black–Scholes greeks vega $\mathcal{V}$, vanna $\mathcal{V}_x$, and volga $\mathcal{V}_u$ (u-derivatives of $C_{\rm BS}$) and $\partial_m^k C_{\rm mix}$ the closed-form strike-derivatives of the mixture price; the last identity annualises each coefficient. These are exact (numerically confirmed against the bisection-inverted IV in App. E); the SSR $\mathrm{SSR} = \partial_z\hat\sigma(0)/\hat\sigma'(0)$ matches the closed-form realised covariance (23) to four decimals. We take (24) as the calibration targets.

Leading order: cumulants. A coarser but interpretable route reads the coefficients off the regime-conditional mixture cumulants $\kappa_n = \partial_t^n L(0)$ of the cumulant-generating function

$$L(t) = \log \sum_l w_l\, e^{t\, d_{f,s,l} + \frac{1}{2}t^2 V_{f,s,l}}, \tag{25}$$

via the Gram–Charlier map $u_0 = \sqrt{\kappa_2}$, $u_1 = \kappa_3/6\kappa_2^{3/2}$, $u_2 = \kappa_4/12\kappa_2^{5/2}$ [11, 12]. By the law of total cumulance $\kappa_3 = \mu_3(d) + 3\,\mathrm{Cov}(d, V)$, the skew's leverage term $\mathrm{Cov}(d, V)$ is explicit and the Appendix D Jacobian is analytic; the SSR itself is the realised covariance (23), not a spot-derivative of these regime-conditional cumulants. The cost is accuracy—this proxy sits 8–9% below the exact level (skew 16–20%)—so the cumulants serve interpretation and identifiability, not the fit.

Roughness consistency, and the SSR as a calibration map. The inverted $\boldsymbol{\theta}$ should respect the static-to-dynamic relation $\mathrm{SSR} \to H + \frac{3}{2}$ when the skew decays as $\tau^{H-1/2}$ [32], with H the Hurst exponent of the volatility process [25, 26]. Read backwards this is a calibration rule: the SSR is the trusted observable, so an observed short-T value fixes the model's skew-dynamics exponent ($H = \mathrm{SSR}^\star - \frac{3}{2}$) rather than leaving it free. The diffusive two-OU kernel sits at $H = \frac{1}{2}$ ($\mathrm{SSR} \to 2$); a market $\mathrm{SSR} \approx 1.5$ reads as $H \approx 0$—the roughness a multi-factor extension must hit, a derived target, not a knob.

6.5 The SSR-consistent hedge ratio

The desk-facing payoff is the hedge. The true delta of a strike-K option carries a smile-response term, and for a *fixed* strike the sensitivity is

$$\Delta_{\rm SSR} = \Delta_{\rm BS} + \mathrm{Vega}\cdot\partial_S\widehat\sigma(K), \qquad \partial_S\widehat\sigma(K) = \frac{\mathcal{S}_T\,(\mathrm{SSR}_T - 1)}{S}, \tag{26}$$

where $\mathcal{S}_T = \partial_k\widehat\sigma$ is the ATM skew. The factor $(\mathrm{SSR}_T - 1)$ combines the SSR's at-the-money response, $\partial_S\widehat\sigma_{\rm ATM} = \mathrm{SSR}_T\,\mathcal{S}_T/S$, with the fixed-strike smile-roll $-\mathcal{S}_T/S$. Hence sticky-strike ($\mathrm{SSR} = 1$) gives $\partial_S\widehat\sigma(K) = 0$ and recovers the Black delta, while sticky-delta ($\mathrm{SSR} = 0$) gives $-\mathcal{S}_T/S$. A Black/sticky-strike hedge omits the $(\mathrm{SSR}_T - 1)$ term and bleeds systematic P&L whenever the realised SSR departs from 1—materially so for SPX at short tenors—while the calibrated kernel supplies the adjustment its calibrated SSR implies. We name this delta *SSR-consistent* rather than "correct": (26) is the analytically correct hedge *for a smile rolling at the calibrated*

SSR. Whether it lowers realised hedging variance against the Black and sticky-strike deltas is a separate, empirical claim. The historical replay of Section 8 finds 27–47% below Black at the money, with the broader multi-moneyness study outstanding.

The hedging error. What (26) buys is the residual it removes. For a Black-delta hedge ($\Delta = \Delta_{\mathrm{BS}}$) of a strike-$K$ option, the step P&L decomposes as

$$\mathrm{d}V - \Delta_{\mathrm{BS}}\,\mathrm{d}S = \underbrace{\mathrm{Vega}\cdot\frac{\mathcal{S}_T(\mathrm{SSR}_T-1)}{S}\,\mathrm{d}S}_{\text{smile-roll (first order)}} + \underbrace{\tfrac{1}{2}\Gamma(\,\mathrm{d}S)^2 - \Theta\,\mathrm{d}t}_{\text{gamma–theta}} + \cdots, \tag{27}$$

the smile-roll term being the value change from the smile moving with spot ($\mathrm{d}\widehat{\sigma}(K) = \partial_S\widehat{\sigma}(K)\,\mathrm{d}S$). It is a *first-order* exposure, systematic whenever $\mathrm{SSR}_T \neq 1$, that inflates the hedging-error variance over many rebalances; the SSR-consistent delta (26) folds it into the spot hedge, leaving the usual gamma–theta. The companion vanna—the spot–vol cross-greek, equally closed-form—hedges the second-order cross: under the SSR the realised spot–vol comovement is $\mathrm{SSR}_T\,\mathcal{S}_T(\,\mathrm{d}S)^2/S$, an exposure the calibrated SSR likewise pins.

6.6 The forward-variance readout (VIX as the SPX instance)

The vol-of-vol is *identified* from a readout of the underlying's forward-variance distribution, and enters the fit only as a soft target. Its *level* is the variance-swap rate—a model-free integral of the underlying's own option strip, hence available for *any* European-option underlying with no separate product—while its *dispersion* (vol-of-vol and skew) is what a volatility-index options market prices directly; for SPX that market is VIX, the sharpest instance. For SPX, VIX_T^2 is 30-day forward variance, $Y_T := \mathrm{VIX}_T^2 = \frac{1}{\tau}\mathbb{E}_T\big[\int_T^{T+\tau} v_u\,\mathrm{d}u\big]$ with $\tau \approx 30/365$, a functional of the latent vol state. A factor with rate κ is damped over the readout horizon by $D(\kappa) = (1-e^{-\kappa\tau})/(\kappa\tau)$; the fast factor is strongly damped and the slow factor barely, so the readout is slow-dominated while short-dated smile curvature is fast-dominated. With $V^{\mathsf{F}}, V^{\mathsf{S}}$ the fast/slow factor contributions to the forward variance—affine readout $Y_T = v_\infty + D_{\mathsf{F}}V^{\mathsf{F}} + D_{\mathsf{S}}V^{\mathsf{S}}$, $D_{\mathsf{F}} = D(\kappa_{\mathsf{F}})$, $D_{\mathsf{S}} = D(\kappa_{\mathsf{S}})$—this separates the timescales:

$$\mathrm{Var}(Y_T) = D_{\mathsf{F}}^2\,\mathrm{Var}(V^{\mathsf{F}}) + D_{\mathsf{S}}^2\,\mathrm{Var}(V^{\mathsf{S}}) + 2D_{\mathsf{F}}D_{\mathsf{S}}\,\mathrm{Cov}(V^{\mathsf{F}}, V^{\mathsf{S}}). \tag{28}$$

Remark 3 (Identification caveat)**.** *The clean "two observables, two unknowns" reading—smile curvature and readout width recover* $\mathrm{Var}(V^{\mathsf{F}}), \mathrm{Var}(V^{\mathsf{S}})$*—holds only once the factor correlation* χ_{fs} *is fixed. Our default* $\chi_{fs} = 1$ *pins the covariance term in* (28) *to* $\sqrt{\mathrm{Var}(V^{\mathsf{F}})\,\mathrm{Var}(V^{\mathsf{S}})}$*, keeping two unknowns; with* χ_{fs} *free there is a third.*

In the discrete kernel the forward variance is regime-determined—$\mathrm{VIX}_T = \sqrt{\bar{v}_k}$ on regime k with probability p_k—so the variance-index law is atomic and its calls are regime sums,

$$C^{\mathrm{VIX}}(K) = e^{-rT}\sum_k p_k\big(\sqrt{\bar{v}_k} - K\big)^+, \tag{29}$$

with $\bar{v}_k$ the forward variance carried by regime k: no PDE, no Monte Carlo.

Remark 4 (Atomic vs. dense regimes, and the VIX smile)**.** *The VIX law in* (29) *is atomic because* $\mathrm{VIX}_T = \sqrt{\bar{v}_k}$ *is a deterministic function of the discrete regime; hence the VIX call is piecewise-linear in* K *and the VIX digital is a step function. This is harmless under our scope, where VIX only* identifies *the regime dispersion, and the atomic sum is a quadrature for that dispersion whose* *accuracy improves with the regime count (a discretisation, not a parameter, knob: the generator stays ~ 6-dimensional). If a smooth VIX smile is required, give each regime a within-regime log-normal spread, whereupon* (29) *becomes a mixture of Black calls—at the cost of one vol-of-VIX parameter per regime.*

The readout enters the objective (21) through a soft target on the kernel's implied vol-of-vol, computed from the closed-form regime sums (29):

$$\mathcal{R}_{\mathrm{vov}}(\boldsymbol{\theta}) = \big(v_\infty(\boldsymbol{\theta}) - v_\infty^\star\big)^2 + \sum_n \big(\xi_{\mathrm{model}}(\tau_n;\boldsymbol{\theta}) - \xi_{\mathrm{VIX}}(\tau_n)\big)^2, \tag{30}$$

the squared mismatch of the model's forward-variance *level* v_∞ and *vol-of-vol* ξ to the readout, summed over the horizons $\{\tau_n\}$ (whose term structure pins κ_{S} through the damping $D(\tau)$; a vol-of-vol-skew term is added where a vol-index smile is liquid). It is *not* a hard marginal: we impose only *unconditional* forward-variance consistency, $\sum_k p_k y_k = \mathrm{FV}$, which is necessary but not sufficient for full joint consistency—the conditional identity $y \approx \mathbb{E}[L_\tau \mid S, Y]$ that the joint problem [30] requires is exactly what reading rather than jointly fitting forgoes.

6.7 Extracting the readout, and what stands in for VIX

The targets ξ_{VIX} and κ_{S} are read by a small, robust estimator rather than a full smile fit, since the readout is a soft target. For SPX the instruments are VIX:

1. **Instruments.** VIX *futures* for the forward-variance level, and VIX *call-spreads*—not pure digitals, as the model variance-index law is atomic (Remark 4)—for the smile width and skew, across the listed maturities $\{\tau_n\}$.

2. **Filtering.** Retain quotes inside liquid bid–ask bands and weight each by inverse spread.

3. **Targets.** Futures $\to$ level / long-run variance v_∞; smile *width* $\to$ vol-of-vol ξ_{VIX}; smile *skew* $\to$ vol-of-vol skew; *term structure* of the width across $\{\tau_n\}$ $\to$ the slow rate κ_{S} via the damping $D(\tau)$.

4. **Map to the generator.** Fix the two timescales $\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$ from the term structure of $\xi_{\mathrm{VIX}}(\tau_n)$ through $D(\tau)$ at two horizons; split $\nu_{\mathsf{S}}, \nu_{\mathsf{F}}$ from ξ_{VIX} and the short-dated smile curvature via (28), conditional on χ_{fs}. All enter $\mathcal{R}_{\mathrm{vov}}$ as soft targets.

For an underlying *without* a liquid volatility-index options market the same recipe runs against whatever readout exists, at graded identification strength (Table 2): the target is never empty, merely sharper for SPX. Off-SPX the readout is *weakly* identified rather than absent: the own-strip skew and curvature give ρ and ξ separately but *fuzzily* (the soft pair of Appendix D), still risk-neutral but noisy. A realised estimate optionally firms it up, at the cost of mixing in a real-world ($\mathbb{P}$) quantity. VIX simply makes the same $\mathbb{Q}$ separation sharp—the honest cost of generality. The vol-of-vol *skew* (the upward VIX-smile slope) is third-order for the SSR and near-ATM dynamics and has no off-SPX market source, so the parsimonious kernel omits it; it is fitted only for VIX-pricing or deep-wing work, where SPX's liquid VIX makes it available.

The CBOE VIX is the log-contract, equal to forward variance only without jumps; a diffusive Gaussian-mixture satisfies this, while a jump component needs the standard convexity correction.

Underlying	Forward-variance readout	Strength
SPX	VIX futures + options	sharpest (market-pinned)
SX5E	VSTOXX futures + options	strong
Crypto (BTC/ETH)	DVOL index + smile + realised	good
NDX, NKY, KOSPI, HSI	vol-index level + smile curvature + realised	workable
FX (EURUSD, ...)	risk-reversal/butterfly + forward-vol	workable
European single names	smile curvature + realised + event structure	weakest
any European underlying	own-strip level + smile curvature + realised	universal fallback

Table 2: The forward-variance readout that identifies the kernel's vol-of-vol, by underlying. The *level* is always recoverable from the option strip; only the *dispersion* (vol-of-vol) varies in identification strength. VIX is the sharpest available instance, not a precondition.

7 Scheduled-event de-eventing

Section 4.8 interpolates *within* a time-homogeneous block. Scheduled events—FOMC and CPI for an index, earnings for a single name—are the block *boundaries*: a known date t_k at which extra variance arrives and breaks the smooth forward-variance level g_0 of Section 4.8. The total Black–Scholes variance to maturity carries a calendar step,

$$w(T) = w^{\mathrm{diff}}(T) + \sum_{t_k \le T} J_k, \tag{31}$$

a smooth diffusive curve plus a sum of event lumps J_k (the saw-tooth in the ATM total-variance term structure). The homogeneous kernel cannot absorb a lump: its per-step conditional variance is $O(\Delta)$ while J_k is an $O(1)$ point mass in calendar time—this *is* the time-homogeneity break. De-eventing strips the lumps so the homogeneous generator $\boldsymbol{\theta}$ calibrates on the clean windows, with the event a thin separate layer; left in, a flexible surface silently absorbs the event premium and the term-structure interpolation can turn calendar-arbitrageable [37, 35].

Two things must be recovered, and the established methods recover only one. Every event-vol construction—a scalar variance bump, a dirty-to-clean surface split, an additive jump density [34, 37]—models the event *marginal*: the before-to-after return distribution at one horizon. None records whether the volatility surface jumps *with* the spot at the release—the event's spot–vol *dynamics*, an event-conditional skew-stickiness ratio. A marginal cannot carry it (it does not pin the response, Theorem 1), so a surface-plus-jump object has nowhere to hold it; a transition kernel does. De-eventing here therefore returns *both* halves—the jump density *and* its spot–vol leverage—and the second, the dynamics, is the part the literature leaves open.

7.1 The clean-flank bridge

Bracket the event window $[T_2, T_3]$ (the one window spanning t_k) by event-free windows $[T_1, T_2]$ and $[T_3, T_4]$—the *clean flanks* that name the method. The marginals $\mu_1, \dots, \mu_4$ are read from the chains; on the clean flanks the kernel is the homogeneous diffusive kernel $\mathcal{K}_{12} = \mathcal{K}_{34} = \mathcal{K}_{\mathrm{diff}}(\boldsymbol{\theta})$, so $\boldsymbol{\theta}$ is identified *where there is no event.* Time-homogeneity then fixes the middle window's diffusive counterfactual—propagate the pre-event marginal through the clean generator,

$$\mu_3^{\mathrm{diff}} = \mathcal{K}_{\mathrm{diff}}(\boldsymbol{\theta})\, \mu_2, \qquad \mathcal{K}_{23}^{\mathrm{obs}} = \mathcal{K}_{\mathrm{diff}}(\boldsymbol{\theta}) * \rho_J, \tag{32}$$

with ρ_J the event-jump law and $*$ convolution in log-return. The event is the gap $\mu_3 \ominus \mu_3^{\mathrm{diff}}$. A pair of marginals alone does not pin the middle kernel; the bridge closes it with two ingredients held fixed—$\boldsymbol{\theta}$ from the clean flanks, and a low-cumulant parametric ansatz for ρ_J.

7.2 Two channels

The event splits along the same statics/dynamics seam as the model.

Statics: the jump density. Take ρ_J *additive*—independent of the diffusive state—so (32) is a genuine convolution and cumulants add. The jump cumulants are then the exact residual of the observed and de-evented marginals,

$$\kappa_n(\rho_J) = \kappa_n(\mu_3) - \kappa_n(\mu_3^{\mathrm{diff}}), \qquad n \geq 2, \tag{33}$$

recovered by a parametric (small-mixture) deconvolution at the digital level—well-posed exactly because ρ_J is low-dimensional, where a non-parametric deconvolution would not be. The second cumulant returns the desk's scalar event variance; the third and fourth carry the event crash-skew and the relief/sell-off bimodality. Writing $\rho_J = \sum_m \pi_m^J \mathcal{N}\big(\mu_m^J, (\sigma_m^J)^2\big)$ with $\sum_m \pi_m^J e^{\mu_m^J + \frac{1}{2}(\sigma_m^J)^2} = 1$ keeps the de-evented and re-evented surfaces in closed-form finite-mixture class.

Dynamics: the event leverage. The marginal μ_3 does not record whether the event jump is *correlated* with the volatility move—a joint spot–vol property, the event-conditional analogue of the SSR. It is read not from μ_3 but from the forward-start smile across $[T_2, T_3]$: the realised covariance (23) evaluated on the event transition, the post-event ATM response $\sigma(\cdot; f', s') - \sigma(\cdot; f, s)$ weighted by the event-step leverage $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$.

7.3 Hedging through the event

An event is a known-time jump, and a delta hedge cannot catch a jump; event risk is therefore an *options* hedge, with a delta correction from the leverage, and the two channels supply both. The static jump sizes the *gamma* leg: its variance is the second cumulant $J = \kappa_2(\rho_J)$, an implied one-event move $\sqrt{J}$ that the bridge separates from the diffusive variance and that sizes the event straddle, while κ_3, κ_4 set the directional tilt—risk-reversals or strangles positioned at the legs of ρ_J. The dynamics supply the *delta* leg: the spot hedge to hold *through* the release is the SSR-consistent delta (26) evaluated with the event-conditional SSR,

$$\Delta_{\mathrm{SSR}}^{\mathrm{ev}} = \Delta_{\mathrm{BS}} + \mathrm{Vega} \cdot \frac{\mathcal{S}_T\,(\mathrm{SSR}^{\mathrm{ev}} - 1)}{S}, \qquad \mathrm{SSR}^{\mathrm{ev}} = \left.\frac{\mathrm{Cov}(\Delta\sigma_{\mathrm{ATM}}, r)}{\mathrm{Var}(r)\,\mathrm{skew}}\right|_{[T_2, T_3]}, \tag{34}$$

the realised covariance (23) taken on the event transition rather than the stationary chain, with vanna the event spot–vol cross-hedge; a Black/sticky-strike hedge sets $\mathrm{SSR}^{\mathrm{ev}} = 1$ and bleeds the $(\mathrm{SSR}^{\mathrm{ev}} - 1)$ term across the release. Holding that delta through the date, the residual attributes cleanly,

$$\Delta\Pi \;\approx\; \underbrace{\tfrac{1}{2}\Gamma(\Delta S)^2 - \Theta\,\Delta t}_{\text{diffusion}} \;+\; \underbrace{\mathrm{Vega}\cdot\Delta\sigma_{\mathrm{ev}}}_{\text{jump }(\kappa_2)} \;+\; \underbrace{\mathrm{Vanna}\cdot\Delta S\,\Delta\sigma_{\mathrm{ev}}}_{\text{leverage }(\mathrm{SSR}^{\mathrm{ev}})}, \tag{35}$$

into a diffusive roll, the event jump, and the leverage cross-term—the event-localized form of the hedging error (27). So the desk receives, per event, the de-evented and re-evented surfaces with the premium isolated, the implied scenario ρ_J, the gamma and event-delta legs, one coherent surface across the date, and the attribution (35). The first two are standard desk de-eventing; the event-conditional delta and the leverage term are what the transition kernel adds over a surface-plus-jump object.

The split is honest about where the value sits. For index macro events, whose priced jump is variance with little directional skew [37], the hedge is dominated by the gamma leg and the event-conditional delta correction is small unless the leverage is non-trivial (the empirical question of Section 8.2); for single-name earnings, whose asymmetry is priced [36], the directional and event-conditional-delta legs carry more weight.

7.4 Localised Gyöngy, and what is and is not new

The bridge is the Gyöngy fusion (17) localised to the event date: the same marginal match, the overlay a calendar lump rather than the full local-volatility surface. Interpolation and de-eventing are one homogeneity property read two ways—the semigroup composes *within* an event-free block (Section 4.8), the lump J_k is the *between-block* residual it cannot bridge. Relative to the established static de-eventing cited at the head of this section, what (32) supplies is the dynamics: an event-conditional SSR that the unconditional-SSR literature does not measure. A full-surface ultra-short model that removes FOMC days notes this accommodation "could be ... but is beyond the objectives of the current paper" [38]. Algorithm 2 collects the clean fit, the counterfactual, and the two channels into one pass.

Algorithm 2 Scheduled-event de-eventing (one event t_k).

Require: marginals on windows bracketing t_k: event-free flanks $[T_1, T_2], [T_3, T_4]$ and the event-spanning window $[T_2, T_3]$; the event calendar

Ensure: the diffusive generator $\boldsymbol{\theta}$, the de-evented smile μ_3^{diff}, the event-jump density ρ_J, and the event leverage

1: **Clean fit.** Calibrate $\boldsymbol{\theta}$ on the event-free flanks only (Algorithm 1 restricted to $[T_1, T_2], [T_3, T_4]$); the event-spanning window is held out.

2: **Counterfactual.** Propagate $\mu_3^{\text{diff}} = \mathcal{K}_{\text{diff}}(\boldsymbol{\theta})\, \mu_2$, the de-evented prediction at T_3.

3: **Statics.** Form the cumulant residual (33) and fit a small martingale-normalised mixture ρ_J (parametric deconvolution); the de-event smile is μ_3^{diff}, the re-evented smile $\mu_3^{\text{diff}} * \rho_J$.

4: **Dynamics.** Evaluate the event-conditional SSR (23) on the $[T_2, T_3]$ transition from the forward-start smile; record the event leverage.

5: **return** $\boldsymbol{\theta}$, μ_3^{diff}, ρ_J, event leverage.

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8 Empirical study: ground-truth validation and real-data calibration

The study has two halves. On synthetic ground truth—where the generator is known—we verify what the market cannot: that the closed forms are exact, that the statics/dynamics decoupling is a genuine bounded degree of freedom, and that the de-eventing estimators recover planted truth (Sections 8.1, 8.2). The real-data study is the application: real SANOS marginals, the SSR+vov fit across regimes, the off-SPX (NDX) generalisation, and the SSR-consistent hedging replay (Section 8.3).

8.1 Ground-truth validation on synthetic data

All checks run at a fixed reference generator $\boldsymbol{\theta}_0$ (nine knobs, calibrated once to canonical SPX term-structure targets; values in Figure 2), on a synthetic arbitrage-free chain where every quantity is computable both in closed form and by simulation.

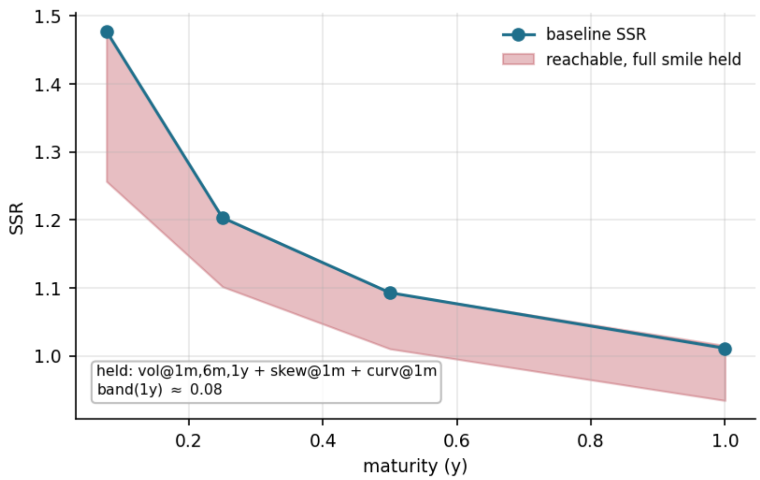

Figure 2: The statics/dynamics decoupling band at the reference generator (closed form, $n_k = 20$). Moving along the steepest *static-preserving* direction—the null-space of the static Jacobian holding *five* smile observables (ATM vol at 1m/6m/1y; skew and curvature at 1m), Newton-held to $\lesssim$ 1 bp in the vols and 10^{-3} in skew—the SSR is freely movable only inside the shaded band: the long end spans $\approx$ 0.93–1.02 (width 0.08) at *fixed* statics. $\boldsymbol{\theta}_0 = (\bar{\gamma}\,-5.30, \nu_{\mathsf{F}}\,0.43, \nu_{\mathsf{S}}\,0.50, \nu_{\mathsf{L}}\,0.14, \lambda_{\text{skew}}\,-1.48, \lambda_{\mathsf{F}}\,0.98, \lambda_{\mathsf{S}}\,1.65, \kappa_{\mathsf{F}}\,1.00, \kappa_{\mathsf{S}}\,2.34)$.

Decoupling and identifiability. The dynamics are a genuine but *narrow* degree of freedom over the statics. Along the steepest static-preserving direction (the null-space of the static Jacobian holding the five smile observables of Figure 2), the SSR(1y) is freely movable only inside a width-**0.08** band $[0.93, 1.02]$ at *fixed* statics. Relaxing the hold to the three-observable smile (ATM vol at 1m/1y, 1m skew) roughly doubles the reach ($\approx [0.89, 1.07]$), which contains the market long-end SSR ($\approx$1.05). The freedom is non-zero (genuine SV, off the local-vol floor) and bounded—wider moves recruit the statics, which the full calibration supplies. The elasticity-scaled Jacobian has effective rank $\approx$7; the closed-form VIX vol-of-vol pins the *timescale* half of the soft direction (a 12× regularisation), leaving the slow leverage λ_{S} for forward-start skew (Appendix D).

Exactness (Monte-Carlo)—the anchor. The closed-form realised SSR sits within Monte-Carlo noise of a direct simulation ($|\text{diff}| \approx 0.005$ at 1m–1y; the fused SLV SSR ≤ 0.007), and the check has teeth—an earlier source-averaged form was off by 0.105 on a skewed target, which the Monte Carlo caught. The dynamic readouts are exact closed forms, not approximations, and this is what the real-data residuals are read against. With the machinery proven exact here, the regime-dependence and two-factor ceiling on real chains (Section 8.3) are properties of the data and the model's scope, not numerical error.

Scope at the short end. The kernel is diffusive ($H=\frac{1}{2}$), so the model SSR rises toward the short-end limit 2; matching SPX's rough sub-month level ($\approx$1.5) needs a rough multi-factor extension ($\text{SSR} \to H + \frac{3}{2}$ [26]).

8.2 Scheduled-event de-eventing: synthetic validation

We validate both channels of Section 7 on the reference generator $\boldsymbol{\theta}_0$, closed form.

The dynamics channel is non-redundant. Inject an event that bumps only the regime-transition leverage $(\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}})$, leaving the within-step return (d, V) untouched. Because $\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}}$ enter only the transitions, which integrate out of the one-step spot marginal, the event-horizon marginal is *exactly* unchanged ($|\Delta\text{var}| = |\Delta\text{skew}| = 0$ to machine precision) while the forward-start SSR moves monotonically with the injected leverage (Figure 3). The event leverage is thus identifiable, yet *provably invisible* to any static de-eventing. It is the non-redundant degree of freedom the dynamics channel adds over the static literature, and it is what licenses reading the real-data SSR *suppression* (Section 8.3: the same curve, on its leverage-down side) as a genuine dynamics signal rather than a static-variance artifact. A symmetric martingale variance jump, meanwhile, reproduces the index-macro "variance, not skew" fact [37] and leaves this leverage signal intact.

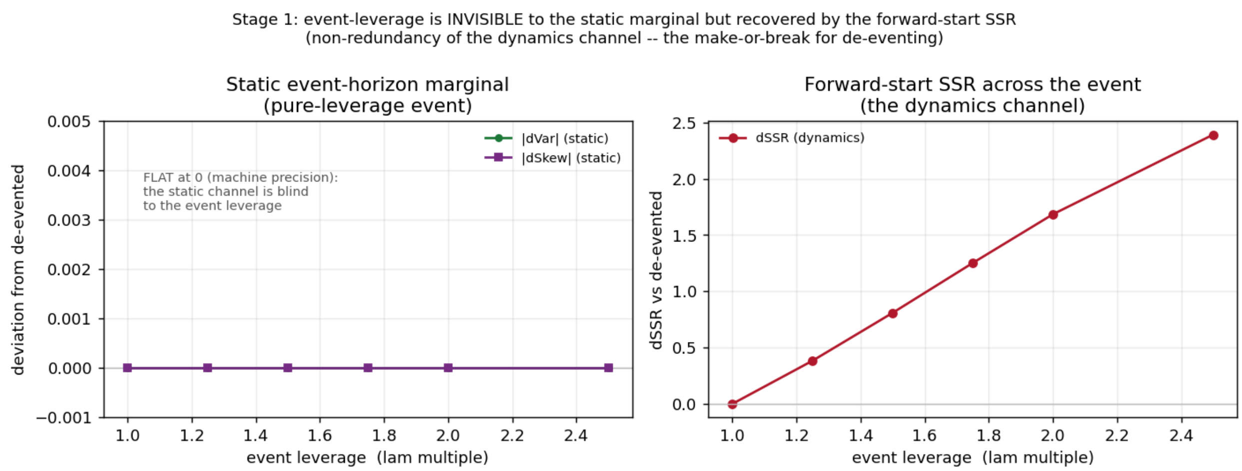

Figure 3: Event-leverage non-redundancy (reference $\boldsymbol{\theta}_0$, closed form). **Left:** the static event-horizon marginal stays at machine-precision zero deviation—a pure-leverage event is invisible to it. **Right:** the forward-start SSR across the event rises monotonically with the event leverage. The dynamics channel carries signal the static marginal provably cannot see.

The static channel recovers a known jump. Injecting a known asymmetric two-leg jump and running the bridge recovers its variance, skew and kurtosis exactly (to $\sim 10^{-16}$) and re-events the smile to 0.00 bp (Figure 4)—the ground-truth check the market cannot give. Because de-eventing is a deconvolution (jump/clean variance 0.38 here), the recovered *variance* is robust ($\pm 4\%$ under $\pm 2\%$ quote noise) while the *skew* is noise-sensitive—itself a mechanism behind the empirical "variance, not skew."

8.3 Real-data study: results

SPX calibration. On real ORATS SPX end-of-day chains the pipeline runs end-to-end. *Statics and propagation:* SANOS marginals fit across $\sim$20 maturities to $\sim$2.5 y (convex order holds), and one admissible kernel propagates between adjacent maturities under the exact martingale lock (13). The marginal-propagation residual $d(\mu_j \mathcal{K}_j, \mu_{j+1})$, after the structural Gyöngy match and re-seeded from each real μ_j, is a median ≈ 3 call-price bp and ≈ 26 ATM-vol bp on the liquid surface (≤ 6 m), within bid–ask. It rises to $\lesssim 35$ call-bp only at the sparse long end, where adjacent expiries are 26–52 steps apart and discretisation accumulates. *Forward-start:* the forward-start return density

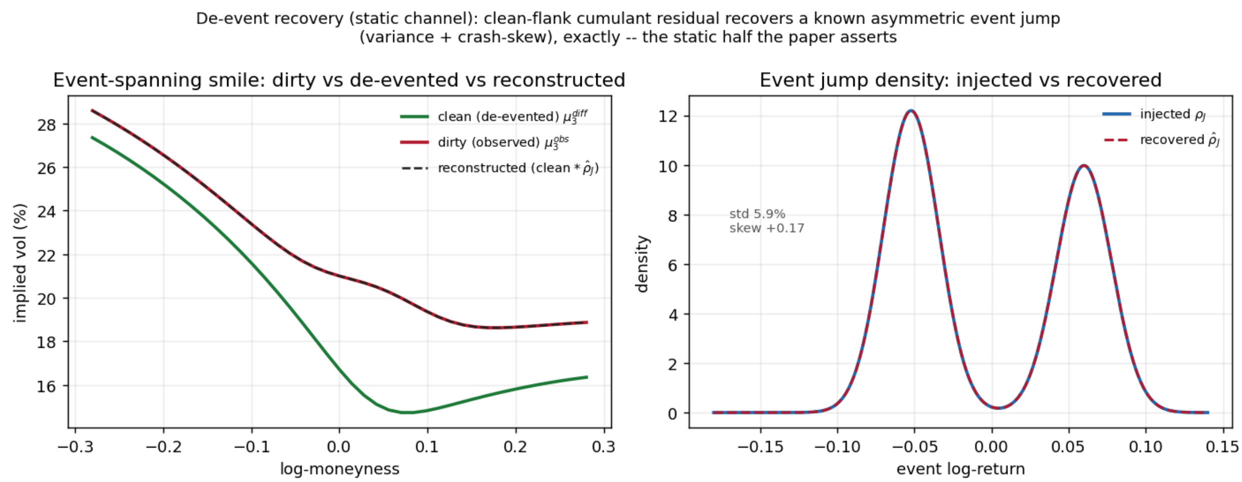

Figure 4: De-event recovery, static channel (3-month window). **Left:** the event-spanning (dirty) smile, the de-evented (clean) counterfactual, and the reconstruction clean $*\,\hat{\rho}_J$, which overlays the dirty smile to 0.00 bp. **Right:** the injected asymmetric two-leg event-jump density and its recovery from the cumulant residual.

is exact (mass and forward equal 1 to machine precision) and its forward skew rolls down with tenor ($\approx -0.5 \to -0.14$), flatter than the spot skew—the smile motion of Section 6. *Dynamics:* the SSR+vov fit reproduces both the realised skew-stickiness term structure (23) and the VIX vol-of-vol, at $\leq$3% RMS on both blocks for calm and moderate dates (Table 3, Figure 5).

Diagnostic	Result (real SPX)	Target
Marginal residual $d(\mu_j\mathcal{K}_j, \mu_{j+1})$	3 call-bp / 26 IV-bp ($\leq$6 m)	within bid–ask
SSR$_T$ fit (calm/moderate)	0.6–2.3% RMS	realised 1.8$\to$1.5
VIX vol-of-vol ξ (soft target)	1% (calm)–9% (high-vol)	VIX-implied
Forward-smile	valid density; skew roll-down	—

Table 3: Real-SPX diagnostics; the multi-start SSR+vov fit over 2015–2023 (autodiff Jacobian, ridge regularisation, cached target; $\sim$250–450 s/date on GPU).

Cross-regime, and the two-factor ceiling. Calm and moderate regimes fit to $\leq$3% on both blocks, while high-vol and post-spike dates expose the two-factor VIX vol-of-vol *shape* limit—its vol-of-vol term structure cannot flatten at the short-mid the way those regimes priced (Table 4, Figure 6a). The *grind* of 2022–2023 is no exception: its realised SSR is normal ($\approx$1.25–1.30), and the 2022 and 2023 SSR+vov fits reach SSR RMS 3.2% and 9.2% (moderate). The residual sits in the VIX vol-of-vol (both $\approx$22%)—the two-factor vov shape ceiling (a vov term structure the two-OU kernel cannot flatten at the short–mid), more severe than in the 2020 high-vol regime (9%), not an SSR misfit. *And the vol-of-vol target does not tax the SSR:* on the 2022 grind, a 25$\times$ increase in the weight on the vol-of-vol term leaves the SSR RMS within 2.3–3.2%, so reading the vol-of-vol off the imperfect VIX target costs nothing on the primary SSR fit. For from-spot outputs the exact second-order leverage (17) is retained: the spot-conditional $\mathbb{E}[\nu \mid z{=}0] \approx 0.7$ (not 1) lifts the accumulated from-spot ATM level by $\sim$10% in vol.

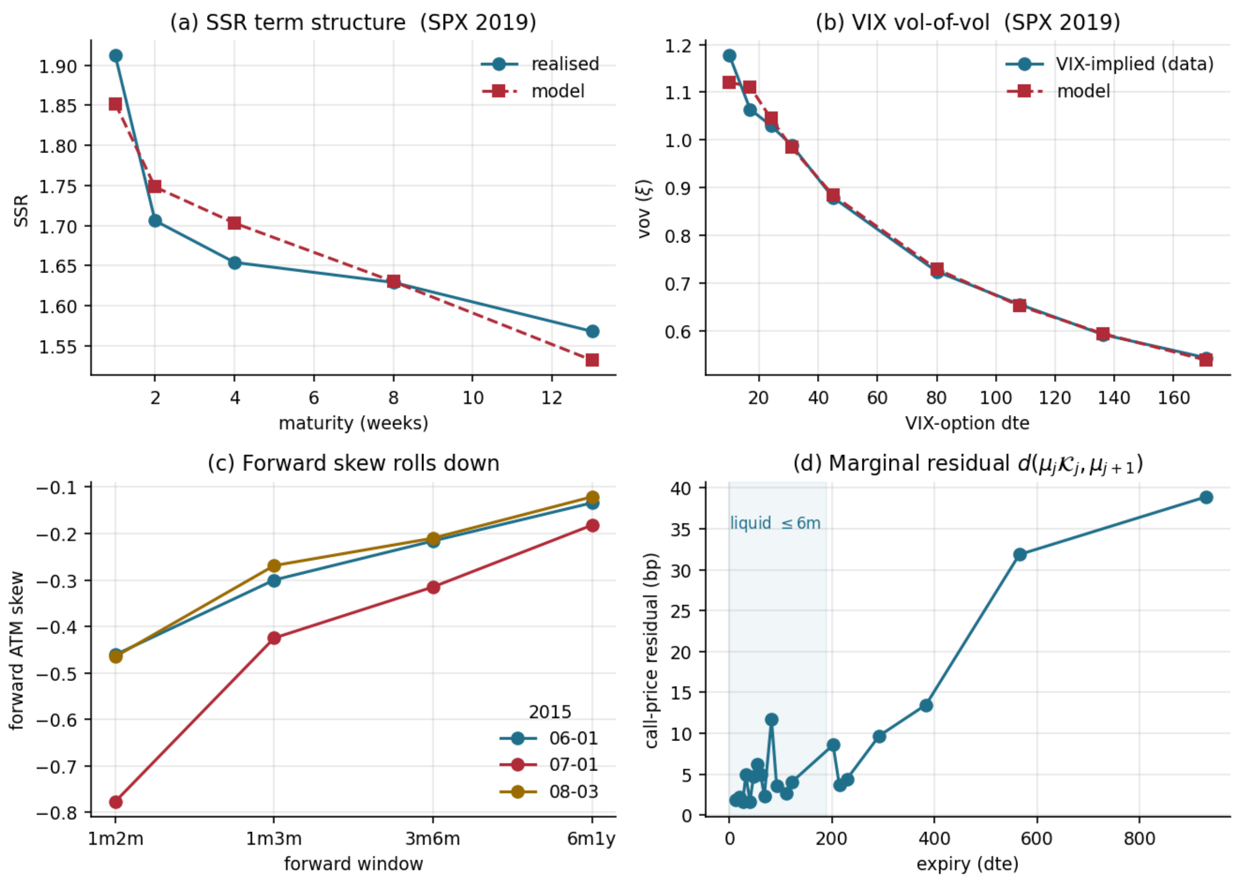

Figure 5: Real-SPX calibration. **(a)** The SSR+vov fit reproduces the realised SSR term structure (2019); **(b)** and the VIX vol-of-vol across VIX-option maturities. **(c)** The forward-start ATM skew rolls down (flattens) with tenor across dates—the smile motion of Section 6. **(d)** The marginal-propagation residual $d(\mu_j \mathcal{K}_j, \mu_{j+1})$ (2015-06-01): a few call-bp on the liquid surface, growing only at the sparse long end where adjacent expiries are many steps apart.

Off-SPX generalisation (NDX). The model travels off SPX with *own-strip and realised targets only*, no vol-index options: NDX statics from its own SANOS marginals, the leverage from its realised SSR, and the vol-of-vol from the realised variance-of-variance of its own model-free variance-swap strip (the graded readout of Section 6; validated in that the reconstructed SPX $\sqrt{V}$ tracks the VIX index). Across 2018–2022, with multi-start, NDX calibrates to 5–12% SSR and 1–8% vol-of-vol (bar the 2021 vov shape ceiling, ∼19%)—weaker than SPX in *both* blocks (its SSR RMS runs roughly double SPX's in matched years, 2018/2020), as expected from the purely physical-measure realised readout (no vol-index, no risk-neutral target) and its ∼10% reconstruction bias. Cross-sectionally NDX's vol-of-vol (∼1.5) exceeds SPX's (∼1.2)—tech is a jumpier vol process—and its marginal is modestly fat-tailed (∼2–3× the data's excess kurtosis), a discretisation feature. The own-strip curvature was tested as a vol-of-vol backbone and kept as a diagnostic, not a fit target (Figure 6).

under.	regime	SSR RMS	VIX / vov RMS
SPX	2015 calm	0.6%	1.2%
SPX	2019 moderate	2.3%	2.6%
SPX	2020 COVID	3.8%	9.0%
SPX	2022 grind	3.2%	22.2%
SPX	2023 grind	9.2%	22.6%
NDX	2018 moderate	8.9%	4.9%
NDX	2021 melt-up	5.0%	18.7%
NDX	2022 bear	6.4%	1.1%

Table 4: Fit quality across regimes and underlyings (SPX: SSR+vov, the vov read from VIX; NDX: SSR+realised variance-of-variance, no VIX). Multi-start GPU wall-time ∼250–450 s/date (SPX), ∼120–190 s (NDX).

Smile dynamics: the level rides, the shape sticks. The same exact-β readout, taken across the whole at-the-money neighbourhood rather than at the level alone, places the calibrated model on the sticky map (the convention of Section 6) in two independent coordinates (Figure 7). The *level* rides with spot at the SSR, $\partial_{\ln S}\sigma_{\mathrm{ATM}} = \mathrm{SSR}\,\mathcal{S}$, with SSR running from 2.0 at one week down to 1.4 at three months: far from sticky-moneyness (SSR=0), between sticky-strike (1) and the local-vol limit (2), and on the Doeff–Kamal $H+\frac{3}{2}$ backbone through the belly. The *shape*, by contrast, barely moves—the skew's own response $\partial_{\ln S}\mathcal{S}$, normalised by the rate at which a strike-pinned smile would swing it (twice the curvature), is only 0.01–0.02: the skew is essentially *frozen in moneyness* (sticky-moneyness *in shape*), with a small, realistic steepening on sell-offs. So the model reproduces the textbook equity picture—the smile *shape* translates with spot while its *level* lifts on sell-offs (Figure 7b: the one-month smile shifts up nearly in parallel on a 1% drop, unlike the flat sticky-moneyness or the tilted sticky-strike responses). This is also why the SSR-consistent delta is the right hedge: with the shape sticky and only the level riding, the whole hedging problem collapses to folding in that level move—the SSR—which is exactly what (26) does.

The frozen shape, however, is as much a *structural constraint* as an empirical match. By translation invariance (Section 4.4) the conditional smile depends on the volatility regime alone, not on the spot level, so the shape can respond to a spot move only *indirectly*—through the leverage-induced regime shift, where a down move tilts the regime toward higher, steeper-skew vol (the small sell-off steepening above). A *deterministic, spot-level-dependent* shape—a smile intrinsically steeper at lower index levels, beyond that stochastic-regime channel—is outside the model's reach:

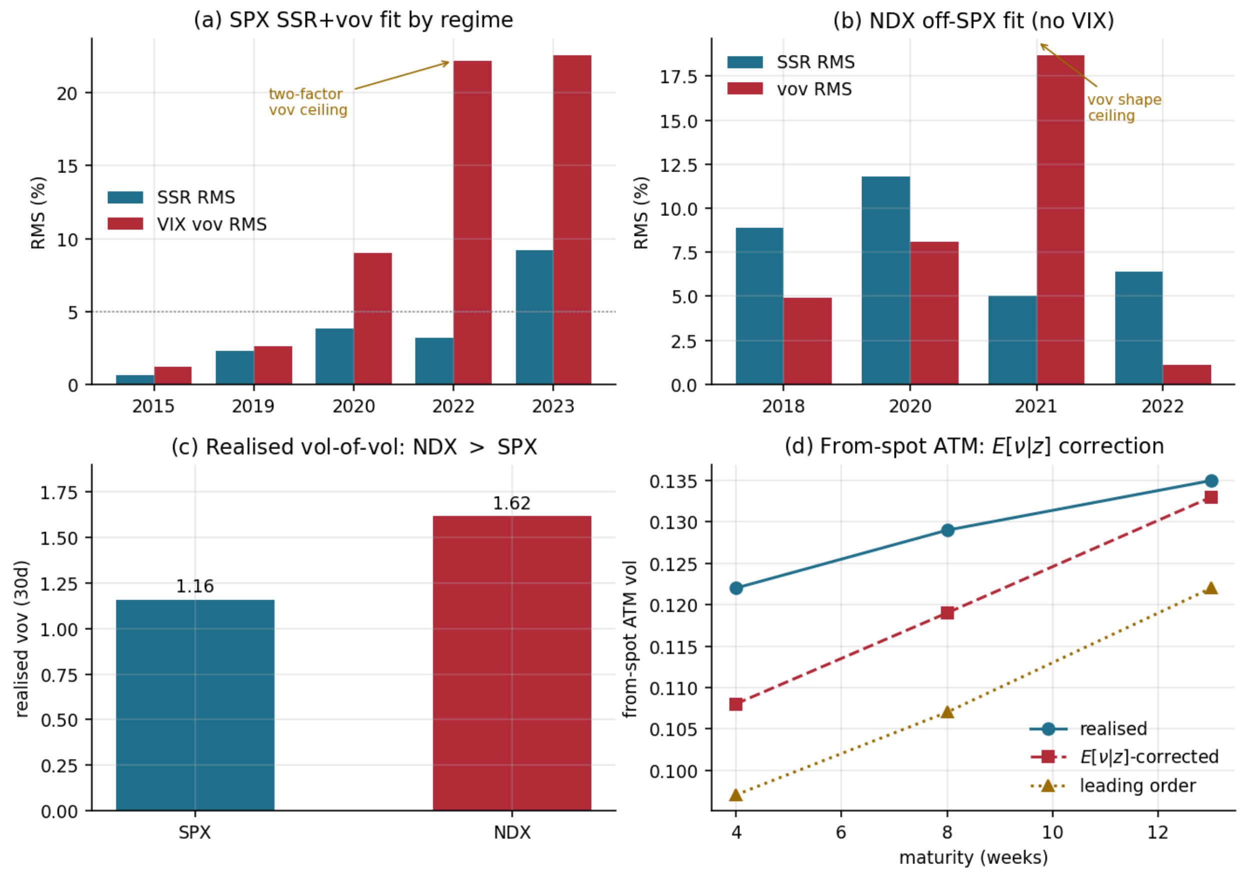

Figure 6: Cross-regime and off-SPX. **(a)** SPX SSR+vov fit RMS by regime—calm/moderate at $\leq 3\%$, high-vol and post-spike (incl. the 2023 grind) exposing the two-factor VIX vol-of-vol ceiling. **(b)** NDX calibrated with own-strip and realised targets *only* (no VIX): 5–12% SSR and 1–8% vol-of-vol except the 2021 shape ceiling ($\sim$19%). **(c)** NDX's realised vol-of-vol exceeds SPX's—tech is a jumpier vol process. **(d)** The second-order $\mathbb{E}[\nu \mid z]$ leverage correction lifts the accumulated from-spot ATM level toward the realised.

the same invariance that makes the SSR an exact closed form forbids it. So where the data carry such second-order, level-dependent skew dynamics, the "shape sticks" finding is a property of the construction as much as of the market.

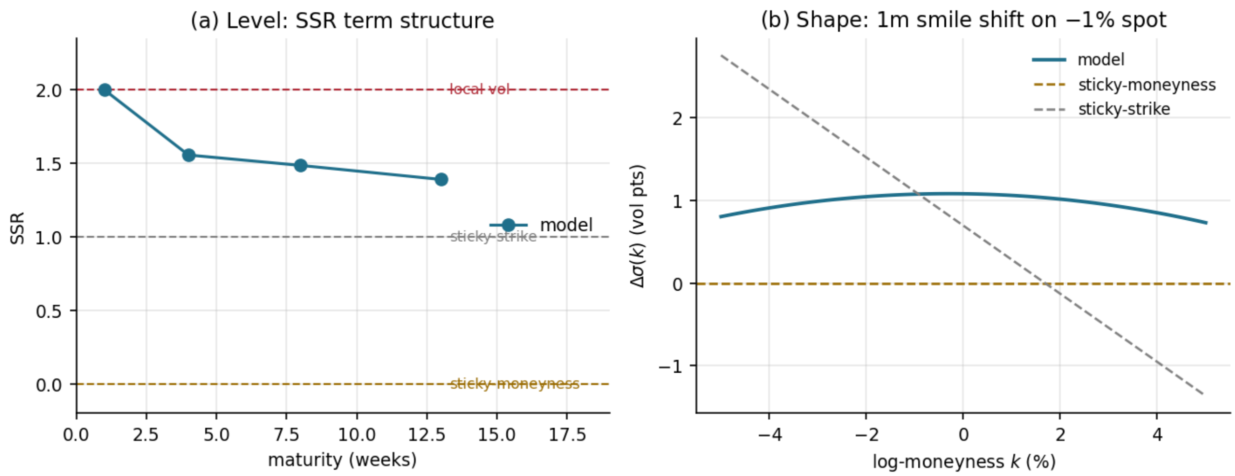

Figure 7: Where the calibrated model sits on the sticky map (SPX, the exact conditional-smile response to a spot move, decomposed by moment). **(a)** *Level*: the SSR term structure runs $2.0 \to 1.4$—between sticky-strike (1) and local vol (2), far above sticky-moneyness (0). **(b)** *Shape*: the one-month smile's response to a -1% spot move is a near-parallel upward shift ($\sim +1$ vol point), close to the flat sticky-moneyness line and unlike the tilted sticky-strike one—the shape is frozen in moneyness while the level rides.

Real-SPX de-eventing, statics: a symmetric variance jump. The clean-flank bridge recovers *both* channels of Section 7, split along the statics/dynamics seam. On the daily-grid era (2022–2024, 48 FOMC+CPI events vs 30 clean controls ≥ 5 days from any release), the counterfactual μ_3^{diff} is a *single* time-rescaled step from the last clean μ_2 (Alg. 2, step 2), taken *without* recompression. A from-spot propagation would instead cold-start the third moment—it builds skew from a point mass over many steps, each recompression shaving it, so at the short event horizon it reads skew ≈ -0.8 against a market ≈ -2.4—whereas anchoring on μ_2 inherits the skew directly. The event jump is then a clean variance lump: the implied move $\sqrt{J}$ ($J = \kappa_2$ of ρ_J) is $\approx 1\%$ on events vs $\approx 0.6\%$ on controls, consistent with the term-structure lump (31), and carries *no* directional skew. Reading the residual asymmetry at *matched* variance (the lump set to the exact κ_2 residual (33), so the two smiles share a second moment and only shape can differ) leaves a large but *event-independent* put-skew deficit: $t \approx 8$ on events and controls alike—the transition-side shape error of a single-step, instantaneous-Gyöngy counterfactual (§4.6, which pins marginals not the transition)—with net event-minus-control $t \approx -0.7$. That is "variance, not skew" [37] on index macro, confirmed and robust to the counterfactual construction.

Real-SPX de-eventing, dynamics: a switched-off event leverage. The signature the static literature cannot form is in the dynamics. For FOMC and nonfarm-payroll releases the event-conditional SSR—the realised $\Delta\sigma_{\text{ATM}} \leftrightarrow r$ slope across the announcement transition (34)—is *significantly suppressed* in normal regimes: across 2015–2019 it falls from a clean ≈ 2.2 to ≈ 0.9 at one week (FOMC dSSR≈-1.4, payrolls ≈-0.7; pooled heteroskedasticity-robust $t\approx3.7$, stable under an $|r|$-crush control)—the smile dropping from near local-vol toward sticky-strike on the release. CPI shows *no* suppression in that low-inflation era (dSSR≈0, $t\approx0.5$); it became the dominant macro surprise only in 2021–22, inside the grind regime below. Calibrated back into the kernel, the event bump reproduces the suppression by driving the stochastic-vol leverage $(\lambda_{\mathsf{F}}, \lambda_{\mathsf{S}})$ to zero, and with it the SSR's stochastic-vol component ($\approx+0.4 \to \approx0$). The event *switches off* the spot–vol leverage rather than adding skew—the kernel-native degree of freedom the marginal provably cannot carry (Section 8.2). The two halves are complementary: a symmetric variance jump the static literature already prices, and a switched-off event leverage only the transition kernel measures.

Hedging: the SSR-consistent delta. The "better delta" claim—the centrepiece—is a historical replay. We delta-hedge a rolling one-month ATM SPX option once per day over the one-parameter family $\Delta(R) = \Delta_{\mathrm{BS}} + \mathrm{Vega} \cdot R\,\mathcal{S}/S$ (26), which spans the standard deltas: R=0 is Black (sticky-strike), $R=-1$ is sticky-delta, and R equal to the SSR is the minimum-variance delta. The one-day hedging-P&L variance is minimised at $R^\star$ equal to the *realised* SSR, and the SSR-consistent delta cuts it sharply below Black (Figure 8, Table 5): across 2016–2021, $R^\star \approx 1.5$–1.8, tracking the realised SSR (1.3–1.6), and hedging at the model's structural R=1.6 reduces the realised hedging variance by **27–47%** versus Black. Sticky-delta ($R=-1$) is *worse* than Black—it moves the delta with the vol–spot comovement the wrong way. Two honest readings temper the headline. That a skew-adjusted delta beats Black on an equity index is the *expected* minimum-variance-delta effect [23, 33]. The replay's content is its size and stability at a fixed, option-implied R. And the replay does not yet isolate the model's value-add over the cheap alternative—a trailing *realised*-SSR delta, which would plausibly track $R^\star$ as well. Whether the model's forward-looking R forecasts next-period comovement better than that backward-looking estimate is the decisive head-to-head, listed below. This edge is a *normal-regime* property: it scales with the realised SSR, so it is largest in calm/moderate markets (SSR $\approx \frac{3}{2}$). The reduction is out-of-sample (a fixed structural R, not re-fit to each hedging window) and survives transaction costs at SPX liquidity: the vol-adjustment term raises turnover, but by a small fraction of the variance it removes, and a smoothed model skew (in place of the raw per-day skew used here) would cut that turnover further. *Still to come:* the full study extends this across moneyness and tenor with tail-loss and cost-curve detail, ablates the VIX layer, the two-timescale memory, and the SSR target, and runs the real-data analogues of the Section 8.1 checks (the Monte-Carlo SSR regression; the identifiability gain of Appendix D on the real readout).

year	days	realised SSR	$R^\star$	var. reduction vs Black
2016	251	1.42	1.70	45%
2017	250	1.28	1.45	27%
2019	251	1.56	1.75	47%
2021	251	1.46	1.60	44%

Table 5: SSR-consistent hedging (one-month ATM SPX, daily rehedge). The variance-minimising delta parameter $R^\star$ tracks the realised SSR; hedging at the model's structural R=1.6 reduces the one-day hedging variance 27–47% versus Black. Sticky-delta ($R=-1$, not shown) *increases* it by 37–46%.

Scope: ATM dynamics; full-smile dynamics as future work. The statics match the full vanilla smile, but the *dynamic* layer is calibrated to *at-the-money* observables—the SSR and the

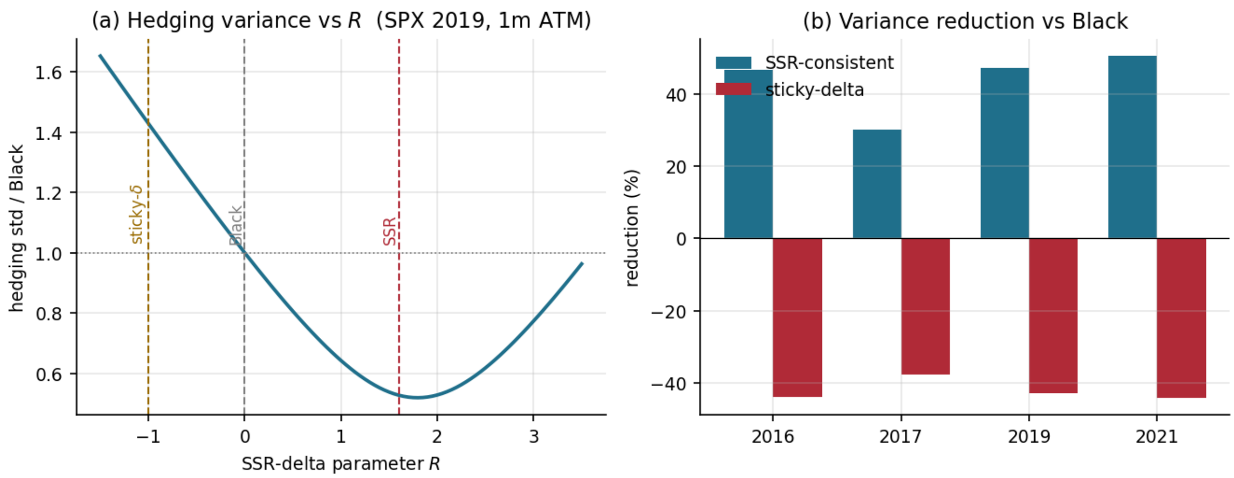

Figure 8: Hedging replay (one-month ATM SPX, daily rehedge). **(a)** The one-day hedging-P&L variance (normalised to Black) as a function of the SSR-delta parameter R, for 2019: a clean U minimised near the realised SSR and far below Black (R=0); sticky-delta ($R = -1$) is worse. **(b)** The SSR-consistent delta (R=1.6) reduces the hedging variance 27–47% versus Black across years, while sticky-delta increases it.

VIX ATM vol-of-vol. Two full-smile extensions are left to a sequel. First, the VIX *smile*—its skew and tails, not only the ATM level—is a richer risk-neutral reading of the vol-of-vol *shape*; the atomic VIX law (Sec. 6) prices it across strikes with no new machinery, tightening the vol-of-vol identification, though—being vol-of-vol—it is *complementary to, not a substitute for*, the leverage that carries the SSR. Second, the *forward-start* smile skew is the clean risk-neutral observable of the spot–vol leverage: it pins the slow leverage λ_S that vol-of-vol leaves soft (Appendix D) and would supersede the realised-SSR (physical-measure) proxy used here—but it requires traded forward-start / cliquet quotes, which are over-the-counter and absent from listed-option feeds (ORATS included), so this route is contingent on data. Together these carry the calibration from ATM- to full-smile dynamics, of interest for cliquet and forward-volatility pricing.

9 Discussion

With the construction and the empirical results in hand, we can be precise about what SANOS-Evolve buys over classical SLV, which already matches vanilla marginals and produces smile dynamics.

The differentiator, previewed in Section 1, is the SSR. The documented rigidity of the calibrated stochastic-volatility family [27, 25] is established for the *pure* SV class, but it transfers to classical SLV: the leverage is *consumed* matching the vanilla smile, so the residual spot–vol dynamics—and hence the SSR—falls to the stochastic-volatility backbone [24, 15], and an SLV on a standard backbone inherits the same band. SANOS-Evolve is itself a discrete SLV, so its escape cannot come from the leverage—it comes from the *backbone*: a discrete regime-transition kernel whose two-timescale innovation leverage shapes the SSR term structure directly while the SANOS LP holds the marginals fixed; the calibration of Section 8 realises this on the empirical SPX term structure.

The differences are *not* approximation power—both are universal at the marginal (SLV through a nonparametric leverage function; SANOS-Evolve through W_1-dense Gaussian-mixture marginals and a sufficiently rich kernel). They are *where the structure sits* and *whether the dynamics are controlled*. Classical SLV sits on the other side of each contribution of Section 1. Its SSR is whatever the stochastic-volatility backbone produces, not a target. Its leverage is a fixed point of the calibration, so re-tuning the dynamics re-triggers it, and it can fail outright when vol-of-vol is large (the leverage $\sigma^2_{LV} = \sigma^2_{\text{Dupire}}/\mathbb{E}[V \mid S]$ diverging), whereas the admissible set here is never empty (Strassen, Theorem 1). Its Monte-Carlo particle noise (worst in the wings, seed-dependent day to day) or PDE grid error sits inside the hedge ratios, where our mixture pricing is deterministic. And its dynamics live in a nonparametric leverage surface, against our nine interpretable knobs and latent regimes.

Universal statics, parsimonious dynamics. The two layers make opposite choices, on purpose. The static layer is universal: W_1-dense Gaussian mixtures (Appendix C) fit *any* arbitrage-free smile exactly—a structural guarantee, not a fitted approximation—so the smile match is never the binding constraint. The dynamic layer is parsimonious: the small propagation residual $d(\mu_j\mathcal{K}_j, \mu_{j+1})$ is a modelling choice, not a wall (a denser kernel, W_1-universal too, shrinks it toward zero), and a few knobs pin the dynamics to interpretable regimes where an unconstrained kernel would leave them underdetermined. Admissibility is exact regardless of the residual (Remark 1: the leverage carries the statics, recompression re-locks the martingale). What lets universality and parsimony coexist is translation-invariance: exact marginal match, few interpretable parameters, and exact closed-form closure—jointly unsatisfiable for a *state-dependent* kernel—hold together because the translation-invariant kernel has exact Gaussian-mixture closure on the parsimonious generator.

Honest costs. The model is discrete, so continuously-monitored barriers need sub-stepping. Recompression keeps the component budget finite (a computational cost, not an approximation limit). The one first-order approximation is the smooth-leverage collocation on the statics, with the dynamics staying exact by translation-invariance. And the framework is empirically unproven relative to battle-tested SLV. Against *frontier* SLV (multi-factor, rough, or neural-SDE [17]), which also fit dynamics flexibly, the edge narrows to the closed-form/deterministic/interpretable combination rather than dynamics control per se. Three further caveats: the empirical edge is regime-dependent—$\leq 3\%$ SSR+vov fits in calm and moderate markets, the two-factor vol-of-vol ceiling in stress (Table 4), which is where dynamics matter most; the two-timescale split is itself the least-identified block (Appendix D); and the static layer rests on SANOS [1], at this writing a preprint. Table 6 is the precise accounting: what is exact, what is approximate, and where each approximation sits.

Outlook: intraday and 0DTE. Same-day-expiry ("0DTE") options are now a majority of S&P 500 option volume, and the construction ports there by *scale invariance*. The marginal mixtures, the kernel, and recompression and interpolation (Secs. 4.7, 4.8) are clock-agnostic, so rescaling the two Ornstein–Uhlenbeck timescales—fast κ_{F} in minutes, slow κ_{S} in hours—re-targets the model at the 0DTE surface. Its natural dynamic target is the *intraday* SSR: by the same $D(\kappa T)$ damping that makes VIX slow-dominated, a minute/hour vol factor is averaged out of any multi-day option and is visible only there. A genuine intraday model needs three localized additions—a time-of-day seasonality $\phi(u)$, scheduled-event martingale jumps (the de-eventing layer of Sec. 7 at intraday resolution), and a terminal/pin refinement as $T \to 0$—plus synchronized intraday NBBO data; we leave the empirical 0DTE study to a dedicated sequel.

status	object	note
exact (closed form)	the translation-invariant kernel and per-fibre lock (§4); the realised-covariance SSR (23); the ATM Taylor coefficients (App. E); the mixture cumulants; static no-arbitrage (Thm. 1, Strassen)	independent of every dynamic approximation; z-invariance is what makes the SSR exact, Monte-Carlo-confirmed ($\lvert\text{diff}\rvert \approx 0.005$)
leverage collocation (check first)	the smooth local-vol leverage $\sigma_{\mathrm{LV}}(z)$ collocated at the component means (§4.6)	moves the statics *level*, not the SSR mechanism (which stays z-exact); single-step benign, but a $\sim$10% multi-month vol-level bias if the exact $\mathbb{E}[\nu \mid z]$ projection (17) is dropped (§4.6)
controlled (convergent)	recompression to $n_{\mathrm{keep}} \approx 24$ (§4.7); the discrete Dupire under-reads at narrow marginals ($\sim$50–85 bp short-T)	accuracy dials—central-difference Dupire, finer grids, tail-clamped leverage
leading-order (cross-check)	the cumulant-to-IV route (§6.4, $\sim$8–9% off)	*not* the primary readout; the exact coefficients (App. E) supersede it
structural limit	short-T diffusive SSR$\to 2$ ($H = \frac{1}{2}$) vs SPX's rough ≈ 1.5	a modelling gap, not numerical error; fix = a rough multi-factor extension
structural limit	spot-level-*independent* smile *shape*: the conditional smile depends on the vol regime, not z (§4.4)	shape responds to spot only via the stochastic-regime channel; deterministic level-dependent skew dynamics are out of reach—the flip side of translation-invariance
out of scope	the wings / Lee moment-index regime	a finite Gaussian mixture is structurally thin-tailed there; ATM by design

Table 6: What is exact and what is approximate. The contrast with a state-dependent (spot-level) kernel is the point: there the quantity to check first is a measure-transport collapse *inside the dynamics*; here the only first-order approximation is a smooth-leverage collocation *on the statics*, the dynamics left exact by translation-invariance.

10 Conclusion

SANOS-Evolve keeps the part of the option-surface problem that SANOS already solves well—arbitrage-free marginals, the local-volatility content—and evolves it into a stochastic-local-volatility model by fitting a martingale transition kernel with a latent two-timescale volatility regime. The marginals stay strictly arbitrage-free, and the kernel supplies the smile dynamics that local volatility gets wrong. The SSR is an exact closed-form realised covariance of return and regime, and the vol-of-vol is identified—not jointly fitted—from a forward-variance readout: VIX for SPX, but recoverable from any European-option surface. The construction needs only European exercise, so it applies to equity indices, FX, and crypto alike; SPX is the worked instance. Everything is closed-form, the no-arbitrage guarantee is a theorem on an admissible set with Strassen existence, and the desk outputs—forward densities, de-event smiles, SSR-consistent deltas—are functionals of one object. The empirical study (Section 8) validates the machinery on synthetic ground truth and reports the real-data calibration across regimes, with a hedging replay in which the SSR-consistent delta cuts realised variance 27–47% below Black. What remains is the broader multi-moneyness, multi-tenor hedging study.

A The SANOS marginal-calibration LP

For completeness, Algorithm 3 states the SANOS static-layer calibration in this paper's notation; the original is [1]. It fits, per maturity, a weights-only Gaussian mixture whose call surface lands within bid–ask and is strictly free of butterfly and calendar arbitrage (positivity → valid density; forward constraint → martingale; convex-order inequality → calendar). The problem is convex—a linear program for a piecewise-linear smoothing penalty $\mathcal{R}$, quadratic for a quadratic one—and feasible within the quote bands by construction. This is the admissible-marginal layer $\{\mu_j\}$ on which Theorem 1 and its convex-order precondition rest.

Algorithm 3 SANOS marginal-calibration LP at maturity T_j (this paper's notation; after [1]).

Require: strikes $\{K_{j,m}\}$ with bid/ask $[B_{j,m}, A_{j,m}]$, mids $C^{\text{mid}}_{j,m}$, weights $\omega_{j,m}$; forward F_j; fixed anchors $\{a_i\}_{i=1}^N$ in $z = \log(S/F_j)$; bandwidth h; previous-maturity fit q_{j-1}

Ensure: weights $q_j \geq 0$ with $\rho_{S,T_j}(z) = \sum_i q_j^i \mathcal{N}(z; a_i, h^2)$, strictly arbitrage-free

1: Build the linear pricing map $(\Pi_j)_{m,i} = \mathrm{BS}\big(F_j e^{a_i+h^2/2},\, K_{j,m},\, h\big)$, the price of basis component i at strike $K_{j,m}$.

2: Solve the linear program over $(q_j, t) \geq 0$:

$$\begin{aligned}
\min \quad & \sum_m \omega_{j,m}\, t_m \;+\; \lambda_h\, \mathcal{R}(q_j) \\
\text{s.t.} \quad & -t_m \leq (\Pi_j q_j)_m - C^{\text{mid}}_{j,m} \leq t_m \qquad (L^1 \text{ fit to mids}) \\
& B_{j,m} \leq (\Pi_j q_j)_m \leq A_{j,m} \qquad (\text{hard bid–ask box}) \\
& \mathbf{1}^\top q_j = 1, \quad \textstyle\sum_i q_j^i\, e^{a_i+h^2/2} = 1 \qquad (\text{normalisation; forward}) \\
& \widehat{C}_j(K_g) \geq \widehat{C}_{j-1}(K_g) \;\; \forall K_g \text{ on a dense grid} \qquad (\text{calendar / convex order})
\end{aligned}$$

3: **return** q_j; the CDF is $G_{S,T_j}(K) = \sum_i q_j^i\, \Phi\big((\log(K/F_j) - a_i)/h\big)$.

B Proof sketch of Theorem 1

Strike no-arbitrage. For each j, μ_j is a probability measure with $\mathbb{E}_{\mu_j}[S] = F_j$; hence $C_j(K) = D_j \int (s-K)^+ \mu_j(\,\mathrm{d}s)$ is convex and non-increasing in K with the correct bounds, which is equivalent to absence of butterfly arbitrage (Breeden–Litzenberger).

Martingale and marginals. Admissibility gives $\mathbb{E}[e^{Z_{j+1}} \mid X_j] = e^{Z_j}$, i.e. the forward-normalised price is a martingale, and $\mu_j \mathcal{K}_j = \mu_{j+1}$ propagates the law, so the chain has one-dimensional marginals exactly $\{\mu_j\}$ and the discounted underlying is a martingale.

Calendar no-arbitrage. A martingale transition implies, by conditional Jensen, that the forward-normalised marginals are increasing in convex order, $\tilde{\mu}_j \preceq_{\text{cx}} \tilde{\mu}_{j+1}$, which is equivalent to absence of calendar arbitrage for the call surface.

Existence. By Strassen's theorem [20, 21], a martingale kernel with $\mu_j \mathcal{K}_j = \mu_{j+1}$ exists iff $\tilde{\mu}_j \preceq_{\text{cx}} \tilde{\mu}_{j+1}$. SANOS enforces this convex order across maturities, so $\mathcal{A}_j \neq \emptyset$. □

C Wasserstein-1 universality of Gaussian-mixture marginals and kernels

That the mixture layer is *effectively nonparametric*—the expressiveness it gives the statics (Section 1)—rests on a denseness fact, which we make precise here *in the topology that governs option prices*. Gaussian mixtures approximate any arbitrage-free marginal, and any continuous admissible kernel, to arbitrary accuracy, with the approximant exactly forward-matched.

The pricing topology is W_1, not weak.* Work in the spot $s = e^z$. The observables are the undiscounted calls $\Lambda_k(\rho) = \langle \rho, (s-k)^+ \rangle$, whose payoff is 1-Lipschitz in s, together with the forward $\langle \rho, s \rangle$. The coarsest topology making all of these continuous is the Wasserstein-1 topology on $\mathcal{P}_1(\mathbb{R}_+)$, the laws of finite first spot-moment; by Kantorovich–Rubinstein and (20),

$$\sup_k \left|\Lambda_k(\mu) - \Lambda_k(\nu)\right| \;\le\; W_1(\mu,\nu) = \int_0^\infty |G_\mu(s) - G_\nu(s)|\,\mathrm{d}s = \left\|C_\mu - C_\nu\right\|_{\dot{W}^{1,1}},$$

the aggregate digital mispricing—*exactly the band-loss the calibration minimises* (Section 5). Weak* convergence is strictly too weak here: the call payoff $(s-k)^+$ and the forward s are continuous but *unbounded*, hence not weak* test functions, so mass escaping to the tail moves prices while the marginal is held fixed against the bounded-continuous C_b. Wasserstein-1 is precisely weak* *plus* the uniformly-integrable first moment that pins the forward and the option values. The martingale constraint is the single linear condition $\langle \rho, s \rangle = F$.

Lemma 2 (Marginal universality in W_1)**.** *For any arbitrage-free marginal $\rho \in \mathcal{P}_1\big((0,\infty)\big)$ with forward $\langle \rho, s \rangle = F$ and any $\varepsilon > 0$, there is a finite Gaussian mixture η (lognormal in s, Gaussian in $z = \log s$) with $\langle \eta, s \rangle = F$* exactly *and $W_1(\eta,\rho) < \varepsilon$.*

Proof. Quantise, mean-preserving. Because $\langle \rho, s \rangle < \infty$, pick M with $\int_{s>M} s\,\rho < \varepsilon/4$ and partition $[0,M]$ into cells $C_1,\dots,C_N$; replace ρ on each cell, and on the tail (M,∞), by an atom at its conditional *barycentre* $s_j = \mathbb{E}[S \mid C_j]$ with mass $w_j = \rho(C_j)$. The atomic η_{a} then has $\langle \eta_{\mathrm{a}}, s \rangle = F$ exactly, and—moving each unit of mass no further than its own cell, the tail cell's cost being $\le 2\int_{s>M} s\,\rho$—

$$W_1(\eta_{\mathrm{a}},\rho) \;\le\; \max_{j\le N} \mathrm{diam}(C_j) \;+\; 2\int_{s>M} s\,\rho(\,\mathrm{d}s) \xrightarrow[\text{mesh}\to 0,\ M\to\infty]{} 0.$$

Mollify, mean-preserving. Replace each δ_{s_j} by a lognormal L_j with mean $\mathbb{E}[L_j] = s_j$ and log-variance σ^2 (a Gaussian in z); the forward stays F exactly, and by convexity of W_1 under mixing,

$$W_1(\eta,\eta_{\mathrm{a}}) \;\le\; \sum_j w_j\,\mathbb{E}|L_j - s_j| \;\le\; \Big(\textstyle\sum_j w_j s_j\Big)\sqrt{e^{\sigma^2}-1} \;=\; F\sqrt{e^{\sigma^2}-1} \xrightarrow[\sigma\to 0]{} 0,$$

using $\mathbb{E}|L_j - s_j| \le s_j\sqrt{e^{\sigma^2}-1}$ (Cauchy–Schwarz). Choosing the mesh and M, then σ, small enough gives $W_1(\eta,\rho) < \varepsilon$ with the forward matched throughout. □

Proposition 3 (Kernel universality in W_1)**.** *Let $\mathcal{K}$ be a martingale kernel on $(0,\infty)$ ($\langle \mathcal{K}(s,\cdot), s' \rangle = s$ for every s) and let $\mu \in \mathcal{P}_1\big((0,\infty)\big)$. For any $\varepsilon > 0$ there is a Gaussian-mixture martingale kernel $\hat{\mathcal{K}}$—each fibre a finite mixture with $\langle \hat{\mathcal{K}}(s,\cdot), s' \rangle = s$ exactly—such that $W_1(\mu\hat{\mathcal{K}}, \mu\mathcal{K}) < \varepsilon$. No regularity of $\mathcal{K}$ is required; and because every fibre keeps its mean, the approximation respects the convex order (Strassen), so the whole SANOS chain $\mu_0 \preceq_{\mathrm{cx}} \mu_1 \preceq_{\mathrm{cx}} \cdots$ is reproduced.*

Proof. Run the Lemma 2 construction on every fibre with one *fixed* grid: cells $C_1, \ldots, C_N$ partitioning $(0, M]$ plus the tail (M, ∞), fibre weights $w_j(s) = \mathcal{K}(s, C_j)$ and barycentres $s_j(s) = \mathbb{E}[Y \mid Y \in C_j;\, s]$—measurable in s for any Markov kernel—each atom mollified by a mean-matched lognormal of log-variance σ^2. Barycentric quantisation keeps every fibre a martingale exactly (no re-centring), and the per-fibre Lemma bound reads

$$W_1\big(\hat{\mathcal{K}}(s,\cdot), \mathcal{K}(s,\cdot)\big) \;\le\; \max_j \operatorname{diam}(C_j) \;+\; 2\int_{y>M} y\, \mathcal{K}(s,\, \mathrm{d}y) \;+\; s\sqrt{e^{\sigma^2} - 1}.$$

Wasserstein-1 is convex under mixing (glue the fibrewise couplings), so integrating against μ—with $\int s\, \mu(\,\mathrm{d}s) = F$ and $\iint_{y>M} y\, \mathcal{K}(s,\, \mathrm{d}y)\, \mu(\,\mathrm{d}s) = \mathbb{E}_{\mu\mathcal{K}}\big[Y\, \mathbf{1}_{Y>M}\big] \to 0$ as $M \to \infty$ (a tail first moment of $\mu\mathcal{K} \in \mathcal{P}_1$)—gives

$$W_1(\mu\hat{\mathcal{K}}, \mu\mathcal{K}) \;\le\; \int W_1\big(\hat{\mathcal{K}}(s,\cdot), \mathcal{K}(s,\cdot)\big)\, \mu(\,\mathrm{d}s) \;\le\; \max_j \operatorname{diam}(C_j) + 2\, \mathbb{E}_{\mu\mathcal{K}}\big[Y \mathbf{1}_{Y>M}\big] + F\sqrt{e^{\sigma^2} - 1},$$

which is $< \varepsilon$ for M large, the mesh fine, and σ small. Convex order is preserved because each fibre has mean s, so $\mu \preceq_{\mathrm{cx}} \mu\hat{\mathcal{K}}$ exactly as $\mu \preceq_{\mathrm{cx}} \mu\mathcal{K}$; composing mean-preserving steps reproduces the increasing-convex-order chain. □

Remark 5. *Two construction points distinguish this from a naive argument. (i) Every step is* mean-preserving*: barycentric atoms and mean-matched lognormal components hold the forward* $\langle \eta, s\rangle = F$ *exactly at each stage, so the approximation never leans on a renormalisation or a global rescaling and positivity is automatic. This is the population analogue of the per-fibre lock* (13)*, which fixes* $\langle \mathcal{K}(s,\cdot), s'\rangle = s$ *exactly at each regime—in both, the martingale identity is enforced by construction, not approached in the limit. (ii) No regularity of* $\mathcal{K}$ *is assumed: the fixed-grid data* $w_j(s) = \mathcal{K}(s, C_j)$ *and the barycentres are measurable for any Markov kernel, so* $\hat{\mathcal{K}}$ *is a genuine kernel with no continuity hypothesis. Our own kernel needs none of this for its* own *convergence—being z-translation-invariant and Gauss–Hermite–discretised it converges to the continuous two-factor model by quadrature (Section 4); the Proposition is what backs the stronger claim that the mixture class matches* any *admissible transition.*

D Local identifiability of the dynamic kernel

The marginals do not pin the kernel, so it is fair to ask whether the generator $\boldsymbol{\theta}$ is identifiable from the data a desk has. Because every observable is closed-form in $\boldsymbol{\theta}$, this is a finite calculation: the elasticity-scaled Jacobian $\tilde{J}_{ij} = (\partial O_i / \partial \boldsymbol{\theta}_j)\, |\boldsymbol{\theta}_j| / |O_i|$ at the calibrated $\boldsymbol{\theta}^\star$, over nine observables (ATM vol @ 1m,1y; ATM skew @ 1m,1y; curvature @ 1m; SSR @ 1m,3m,6m,1y), has singular values

$$\sigma = (5.05,\, 2.96,\, 1.09,\, 0.59,\, 0.28,\, 0.19,\, \mathbf{0.059},\, \mathbf{0.011},\, \mathbf{0.000}). \tag{36}$$

Seven stiff directions, then a gap to two near-flat ones: **effective rank** $\approx 7/9$ at a 1% threshold (full algebraic rank, but condition number $\sim 10^4$). The softest singular vector loads on $\lambda_{\mathsf{S}}, \kappa_{\mathsf{S}}, \lambda_{\mathsf{F}}, \nu_{\mathsf{F}}$—the *SSR-shaping subspace*: the four leverage/timescale knobs that let the kernel fit the SSR are not separately identified by a single date's smile and SSR term structure, because the SSR curve is too smooth to resolve the fast/slow split (this is *not* the skew $\rho\xi$ pair, which here stays stiff). The level, the vol-of-vol/curvature, the skew, and the SSR *level* are well-pinned; the fast-vs-slow SSR *decomposition* is the soft direction—the honest cost of the model's flexibility, and the price of a *right* SSR. (The spot-coupled six-knob variant was better-conditioned, cond ≈ 19, but only because its pinned long-end SSR was the wrong shape.)

Resolving the soft direction. To pin the fast/slow split, add a dynamic instrument the SSR term structure alone cannot carry. The closed-form VIX vol-of-vol term structure—VIXvov(τ), the stationary-π dispersion of the τ-forward vol, a finite sum over the regime grid—does most of it: adding VIXvov at 30 and 90 days to the nine observables drops the condition number from $\sim 10^4$ to ~ 840 (a 12$\times$ regularisation, σ_9 up 15$\times$). It resolves the *timescale* half (the κ, ν block) but is structurally blind to the leverages (forward-*variance* dispersion does not see λ), so the residual soft direction is the slow leverage λ_{S}, which forward-start skew—not vol-of-vol—must pin. The two dynamic instruments are complementary, not interchangeable.

Decoupling: the positive face. The same soft direction is the statics/dynamics *decoupling*. The directions that leave the smile fixed are $\ker J_{\text{stat}}$, and along them the SSR still moves, at the rate $\|P\,\nabla_{\boldsymbol{\theta}}\text{SSR}\|$ with $P = I - J_{\text{stat}}^{+} J_{\text{stat}}$ the static null-space projector. The resulting band at *fixed* statics ($\approx [0.93, 1.02]$) is reported in Section 8.1 (Figure 2).

Two limits. The statement is *local*, and it settles identifiability of the parameters *if* the market has this structure—not whether the kernel *fits* the market (fit quality, Section 8). Off-SPX, with no volatility-index options, the timescale split is separated only weakly—by the chain's own curvature and a realized estimate—so the (λ, κ) block is the fragile part when the readout is poor; where data is thin, regularise it (a prior on the timescales) rather than floating all nine knobs.

E Closed-form ATM Taylor coefficients by implicit differentiation

This appendix derives the exact ATM implied-volatility coefficients used in Section 6.4 and writes them as explicit component sums; we verify them numerically at the end. (The same implicit-differentiation expansion appears in the GLIDE notes of Cao–Huang, unpublished.)

Let the conditional forward-return law be a Gaussian mixture $\mu = \sum_k \omega_k \mathcal{N}(m_k, v_k)$ in $z = \log(S_T/F)$, martingale-normalised so the forward is one: $\sum_k \omega_k F_k = 1$, $F_k = e^{m_k + \frac{1}{2} v_k}$. Work in log-strike $x = \log K$ and total volatility $u(x) = \widehat{\sigma}(x)\sqrt{T}$. The call price is the finite sum $C_{\text{mix}}(x) = \sum_k \omega_k\ \text{Call}(F_k, e^x, \sqrt{v_k})$ and the implied vol solves the price-matching identity

$$C_{\text{mix}}(x) = C_{\text{BS}}\big(x, u(x)\big), \qquad C_{\text{BS}}(x, u) = \Phi(d_+) - e^x \Phi(d_-), \quad d_\pm = -\frac{x}{u} \pm \frac{u}{2}. \tag{37}$$

Write $\delta_k := m_k/\sqrt{v_k}$ (the ATM d_- of component k), with φ, Φ the standard-normal density and CDF.

Level. At $x = 0$, $C_{\text{BS}}(0, u) = 2\Phi(u/2) - 1$, so the ATM total vol is one probit of the closed-form ATM mixture price:

$$u_0 = 2\,\Phi^{-1}\Big(\tfrac{1}{2} + \tfrac{1}{2} C_{\text{mix}}(0)\Big), \qquad C_{\text{mix}}(0) = \sum_k \omega_k \big[F_k\, \Phi(\delta_k + \sqrt{v_k}) - \Phi(\delta_k)\big]. \tag{38}$$

Skew and curvature. Differentiating (37) in x, with the total-vol Black greeks vega $\mathcal{V} = \partial_u C_{\text{BS}}$, vanna $\mathcal{V}_x = \partial^2_{xu} C_{\text{BS}}$, and volga $\mathcal{V}_u = \partial^2_u C_{\text{BS}}$, gives $\partial_x C_{\text{mix}} = \partial_x C_{\text{BS}} + \mathcal{V}\, u'$ and $\partial_x^2 C_{\text{mix}} = \partial_x^2 C_{\text{BS}} + 2\,\mathcal{V}_x\, u' + \mathcal{V}_u\, (u')^2 + \mathcal{V}\, u''$, which solve for $u' = u'(0)$ and $u'' = u''(0)$:

$$u' = \frac{\partial_x C_{\text{mix}} - \partial_x C_{\text{BS}}}{\mathcal{V}}, \qquad u'' = \frac{\partial_x^2 C_{\text{mix}} - \partial_x^2 C_{\text{BS}} - 2\,\mathcal{V}_x\, u' - \mathcal{V}_u\, (u')^2}{\mathcal{V}}. \tag{39}$$

Using the Black identity $F_k\varphi(\delta_k+\sqrt{v_k})=\varphi(\delta_k)$ to cancel the density terms, the mixture strike-derivatives at $x=0$ are the explicit sums

$$\partial_x C_{\mathrm{mix}}(0) = -\sum_k \omega_k\,\Phi(\delta_k), \qquad \partial_x^2 C_{\mathrm{mix}}(0) = \sum_k \omega_k\Big[\frac{\varphi(\delta_k)}{\sqrt{v_k}} - \Phi(\delta_k)\Big], \tag{40}$$

and the Black reference at $(x{=}0, u_0)$ is

$$\begin{gathered}\partial_x C_{\mathrm{BS}}(0) = -\Phi\big(-\tfrac{u_0}{2}\big), \quad \partial_x^2 C_{\mathrm{BS}}(0) = -\Phi\big(-\tfrac{u_0}{2}\big) + \frac{\varphi(u_0/2)}{u_0}, \\ \mathcal{V} = \varphi\big(\tfrac{u_0}{2}\big), \quad \mathcal{V}_x = \tfrac{1}{2}\varphi\big(\tfrac{u_0}{2}\big), \quad \mathcal{V}_u = -\tfrac{u_0}{4}\varphi\big(\tfrac{u_0}{2}\big).\end{gathered} \tag{41}$$

Substituting the mixture sums (40) and the Black greeks (41) into (39) gives the total-vol skew $u_1 = u'$ and curvature $u_2 = u''$ in closed form; the annualised ATM coefficients are $\big(\widehat{\sigma}(0), \widehat{\sigma}'(0), \widehat{\sigma}''(0)\big) = (u_0, u_1, u_2)/\sqrt{T}$, with $\widehat{\sigma}'(0) = \mathrm{d}\widehat{\sigma}/\,\mathrm{d}\log K$ the SSR denominator (Section 6). Section 6.4 writes the same expansion in log-moneyness $m=-x$, where $\partial_m = -\partial_x$ flips the odd-order derivative.

Exactness (verified). Equations (38)–(41) are exact—no Black–Scholes inversion, only the probit Φ^{-1} for the level. On random martingale Gaussian mixtures they reproduce the bisection-inverted Black–Scholes implied vol and its finite-difference strike-derivatives to within $\sim 10^{-6}$ for the level and skew, and to finite-difference truncation for the curvature.

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